

CHAPTER 12

LOCATING SIMULTANEOUSLY CREATED BOUNDARIES

12.1 INTRODUCTION

The area of simultaneously created boundaries has been, and still is, a fruitful area for professional surveyor examinations and litigation. It has been found that both state registration questions and court decisions have inadvertently misapplied basic principles that surveyors must understand in order to practice in the daily world of boundary retracement.

Prior chapters have covered the one type of parcel creation—sequential. This is one of junior/senior creation and rights. The second form of real property interest creation is a simultaneous creation. This could be likened to the *big bang* theory whereby there was nothing originally and then, upon the occurrence of certain required events, *it becomes a fact*.

Definition. A simultaneously created boundary results when several smaller parcels of land are created from one large parcel or from several combined large parcels at the same legal instant by the same person, persons, or agency and by the same instrument. All parcels have equal standing, and no such portion can be said to have prior rights or seniority over any other portion.

This concept of simultaneously created boundaries falls into three separate and distinct areas: those created under federal laws (the GLO system), those created under state laws, and those created under combinations of metes and bounds descriptions, wills, and partition proceedings.

The following relevant principles are discussed in this chapter:

- PRINCIPLE 1. The boundaries and corners of lot lines within a subdivision are determined by the exterior lot lines of an older subdivision or by the boundary lines of a metes and bounds description.

- PRINCIPLE 2. The micro boundaries of the interior lot lines located within the senior macro lines should not extend beyond the macro boundaries of the parent parcel.
- PRINCIPLE 3. In a subdivision that is a product of a replat, the retracing surveyor should consult both plats and then realize different rules may apply under certain conditions
- PRINCIPLE 4. Lost corners on exterior (macro) boundaries cannot be retraced from micro corners found on inside lots. These micro corners can be used as collaborative evidence with other evidence to prove obliterated corners but not to proportion from for lost corners.
- PRINCIPLE 5. A subdivider who subdivides another's land cannot convey good title to land improperly monumented. After a period of time, however, a person may acquire title to an improperly monumented strip of land through unwritten methods of title transfer.
- PRINCIPLE 6. A subdivider who describes the boundaries of a subdivision incorrectly but owns all the lands monumented conveys title to the land described improperly.
- PRINCIPLE 7. If interior lots of a subdivision were originally created by protraction, the lots are located by measurements from the subdivision boundaries; if interior lots of a subdivision were created originally by survey with monumentation, the original survey and monuments control lot locations.
- PRINCIPLE 8. The intentions of the parties to a subdivision are paramount to all other considerations.
- PRINCIPLE 9. Once a lot, street, or block line within a subdivision is established by the original surveyor and the land is sold in accordance with the original plat, the lines originally marked and surveyed are unalterable except by resubdivision. Within the subdivision, boundary lines may be an exception.
- PRINCIPLE 10. In a lot and block description, subdivision monuments called for on the plat, or monuments set by others and known to perpetuate the position of the original monuments called for, if properly identified and undisturbed, control the position of the original lot lines.
- PRINCIPLE 11. Monuments other than the original monuments or replacements of the original monuments may become title monuments by prescription, agreement, estoppel, or other means; however, these monuments cannot be used to determine original lines of the subdivision or to control such lines.
- PRINCIPLE 12. Original monuments set on the ground, except where the intent is clearly otherwise, control facts given on a plat.
- PRINCIPLE 13. After due allowance has been made for weathering, deterioration, and other disruptive forces, the monument and its markings should

be substantially the same as the record describing the monument. In the absence of visual evidence of a monument, location of a former monument can be determined by competent witnesses who saw the monument and remembered its location.

- PRINCIPLE 14. Within a subdivision, distance, direction, and area are presumed subordinate to the intent of the subdivider; the lines are as marked and surveyed, and original monuments control. Where excess or deficiency is found, it is distributed in proportion to linear measurements between original found monuments.
- PRINCIPLE 15. Natural monuments, where called for, are presumed to control street lines.
- PRINCIPLE 16. After natural monuments, artificial monuments that represent the actual lines run by the original surveyor at the time of making the plat are presumed to control street lines irrespective of whether the courses, distances, and street improvement agree with the plat.
- PRINCIPLE 17. In the absence of natural monuments or evidence of lines actually run by the original surveyor, improvements, such as curbs and paving, which were installed simultaneous to the original platting in accordance with the original survey monuments, may be presumed controlling.
- PRINCIPLE 18. In the absence of artificial monuments and evidence of the lines as marked and surveyed, where a street is plotted as being the continuation of a nearby street or beginning at a nearby street, the line of the nearby street is presumed to control.
- PRINCIPLE 19. In the absence of evidence covered by the foregoing principles, the exact width of the street as given on the plat and the distances and angles are presumed to govern street location. Occasionally, a measurement index is applied.
- PRINCIPLE 20. If the street width is not given on the plat, the width as scaled on the map will govern as a last resort.
- PRINCIPLE 21. Offset monuments set by the city engineer or other officials to perpetuate the position of the original monuments of the original surveyor control street lines. Offset monuments not based on adjacent original monuments are afforded control only in proportion to the accuracy with which they were set in accordance with the foregoing rules. City engineers' monuments long acquiesced to are presumed to be correct; the contrary must be proved.
- PRINCIPLE 22. Excess or deficiency existing in a straight line between fixed monuments within a subdivision is distributed among all the lots along the line in proportion to their record measurements.
- PRINCIPLE 23. Excess or deficiency occurring within a block should not be prorated among other blocks.

- PRINCIPLE 24. Excess or deficiency existing in a straight line between fixed monuments within a subdivision is distributed among all the lots along the line in proportion to their record measurements.
- PRINCIPLE 25. Excess or deficiency existing between fixed monuments on a curve is distributed among the lots along the curve in proportion to their record linear measure.
- PRINCIPLE 26. Excess or deficiency cannot be distributed beyond any undisturbed original monument.
- PRINCIPLE 27. When the end lot measurement is not given, all the excess or deficiency is presumed to be given to the end lot.
- PRINCIPLE 28. Where excess or deficiency is given to an irregularly shaped lot at the end of a block, the method is called the remnant rule. Few jurisdictions accept it.
- PRINCIPLE 29. In the absence of physical evidence on the ground and in the absence of measurements given on the plat, as a last resort scaling of the plat may be used.
- PRINCIPLE 30. Where the boundaries of a subdivision were staked correctly originally, the foregoing principles for the establishment of lots are applied.
- PRINCIPLE 31. Lots abutting on a subdivision boundary line cannot extend beyond the title interest of the original subdivider except for a right of title resulting from lawful possession.
- PRINCIPLE 32. Where the original subdivider failed to subdivide all the land as shown by found original monuments, the surveyors should not extend the lot lines beyond the limits of the original monuments. Title to the unsubdivided land probably remains in the original subdivider.
- PRINCIPLE 33. A considerable excess or deficiency existing in blocks adjoining a subdivision boundary is prorated only as a last resort.
- PRINCIPLE 34. It is safer to locate lots from the interior of a subdivision than from the boundary line of a subdivision.

12.2 DEFINING SUBDIVISIONS

To have a need for determining a simultaneous boundary, the primary requirement is that there first must be a subdivision being divided into two or more parcels created from a parent parcel, with the following conditions being fulfilled:

1. All the divisions of land parcels are indicated and shown on a map or plat.
2. The resulting map or plat is approved and filed with or by the agency that either created the parcels or by whomever is responsible for compliance with the respective subdivision laws. This will not apply if there is no requirement for recording.

3. The parcels of land are sold by reference to the map or plat, and no parcel is sold or conveyed until the map is either approved or filed.
4. If the subdivision map is not being recorded, then it must be specifically mentioned in the conveyance documents for the information to be controlling,

This order may also apply if the jurisdiction does not require recording of the map or plat. If lots are sold in accordance with a plan or map, and no reference to the lot exists other than the map, no one lot can be said to be senior to any other lot. All lots being created at the same moment of time—that is, at the moment of filing the map or at the moment of the first transfer of title—are of equal standing. A deficiency in a block found to exist within a platted subdivision is divided among the several lots in proportion to frontages as indicated on the plat, without regard to the sequence of their sale by the proprietor.¹ But if lots are sold in sequence and the lots are added to the map after they are sold, senior rights may exist. Locating subdivision lots on the ground presents essentially the same legal elements as those presented for metes and bounds descriptions, except for certain modifications caused by an absence of senior rights. In a metes and bounds description, the owner of the senior title receives all that is coming to him or her, and the junior title holder has the remainder. Within a subdivision, any excess or deficiency is divided among several lots in accordance with principles that have evolved from case law. Excepting federal subdivisions, lands subdivided within a state are governed by the state laws, and relocation of boundaries must be done in accordance with the rules or laws of the state. Lot and block conveyances calling for a plat offer a minimum of written language on the face of the deed; however, this does not necessarily mean that a lot and block description is the simplest conveyance to establish on the ground. Many older subdivisions and some poorly surveyed modern subdivisions present difficult situations to the person attempting to reestablish the true deed lines. Litigation arising from ambiguous lines and figures on maps has produced many principles of common law. The exterior boundaries of a subdivision may or may not have senior conveyance considerations, depending on the original deed of the subdivider. If the deed defining the boundary of the subdivision is junior to the adjoiner, all lots adjoining the senior deed are junior in character to the adjoiner. Lots within a subdivision may be junior to an adjoiner of the subdivision but not to another lot within the same subdivision.

SUBDIVISION BOUNDARIES AND CORNERS

12.3 ALIQUOT PART SUBDIVISION

Principle 1. *The boundaries and corners of lot lines within a subdivision are determined by the exterior lot lines of an older subdivision or by the boundary lines of a metes and bounds description.*

In attempting to retrace subdivision boundaries without relying on stakes that were set for the original subdivision, the surveyor must examine the original

description of the subdivision and the lots within the subdivision proper. This actually is a two-step process. There have been times when the creating surveyor set monuments intending them to identify the corners of the subdivision and the lots. An example of this is in a subdivision where boundaries and/or corners represent an aliquot part of a section. Figure 12.1 shows that the original surveyor who created the exterior boundaries of the subdivision was in error when he set the exterior boundaries, apparently because the surveyor did not properly conduct a true and correct aliquot part survey.

Monuments were set and the respective lots located within the erroneously located subdivision. Figure 12.1 shows that the creating surveyor failed to correctly locate the east–west centerline. Apparently, the original subdivision corner was a “stubbed corner” at 1320 feet (not 20 chains GLO). This scenario presents two valid possibilities or solutions to this problem. The ultimate answer depends on how the descriptions of the subdivision and the lots were written in the subsequent deeds.²

The surveyor must understand the relationships in order to determine the correct solution. The two solutions are:

1. The subdivision and respective lots refer to aliquot portions only and reference to lots, without reference to the monuments set.
2. The subdivision and respective lots refer to the monuments set in the survey of the subdivision.

12.4 CONTROLLING BOUNDARIES

Principle 2. *The micro boundaries of the interior lot lines located within the senior macro lines should not extend beyond the macro boundaries of the parent parcel.*

When the exterior macro boundaries of a parcel are created, they become the limiting perimeters of the individual interior lots

This principle should have great significance with the retracing surveyor in the retracement of boundaries. It requires a basic understanding of the sequential application as well as the simultaneous creation in applying the reasoning. First, the exterior boundaries of the parcel control all lots within the boundaries. With this in mind, if one of the sequential interior corners of a boundary is dependant on an exterior boundary, the exterior boundary should be located first; then the interior corner (boundary) is identified.

Principle 3. *In a subdivision that is a product of a replat, the retracing surveyor should consult both plats and then realize different rules may apply under certain conditions.*

The law permits a land owner to prepare new plats of former platted subdivisions as long as former vested rights are not violated. The replat cannot change any of the former lots that had already been conveyed. If a surveyor is asked to undertake such a project, he/she should approach it with caution.

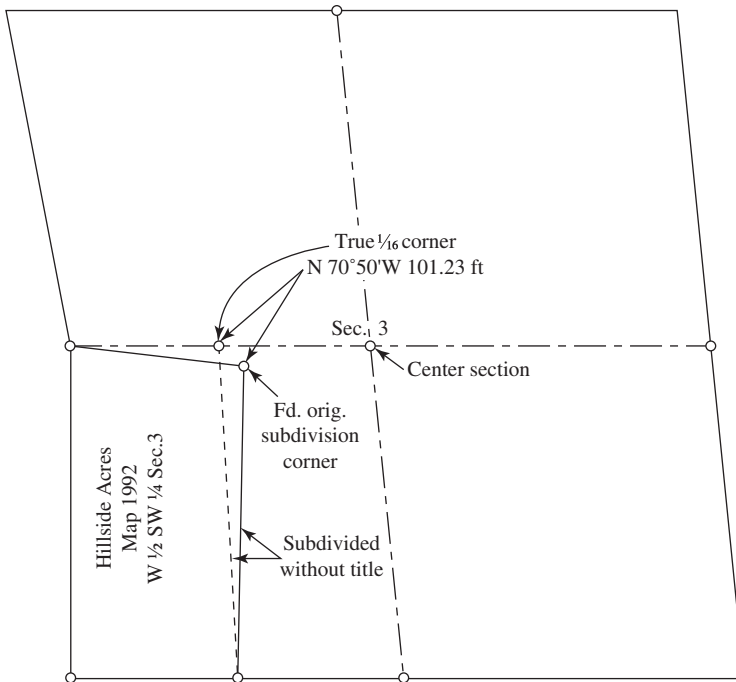


Figure 12.1 Erroneously located subdivision. When the original map, 1992 Hillside Acres, was recorded, the northwest corner indicated a “2-inch iron pipe set at a fence corner in a boundary of long standing.” Upon retracing the original subdivision boundaries, the situation depicted in the figure resulted. In locating the west half of the southwest one-quarter of Section 3, the following was found: an area of land beyond the boundaries of the west half was included in the description to the monuments. Now the surveyor must conduct additional research to determine if the original subdivider owned the east half; that is, was there common ownership of the entire parcel? If this was the case, one avenue for solving the problem could be followed; if not, a second solution would be followed. Without reference to the monuments and with only a reference to an aliquot subdivision, the description controlled. The lot owners who fell outside the east boundary of the west one-half had title problems and possibly could be evicted if the original subdivider did not own the east half. Actions for breach of warranty would be proper. However, if the original descriptions called for set monuments and the original landowner did not own the east half, affected landowners could consider a quiet title action under color of title. If the original subdivision owner owned the east half, there could possibly be a title problem involving junior-senior interest. It is suggested that any surveyor who finds such a situation as this should instantly consult legal counsel and prepare a detailed report and analysis for the client before attempting to undertake any solution.

Principle 4. *Lost corners on exterior (macro) boundaries cannot be retraced from micro corners found on inside lots. These micro corners can be used as collaborative evidence with other evidence to prove obliterated corners but not to proportion from for lost corners.*

Corners placed along exterior boundaries control those boundaries, and the corners of the individual subdivision lots are separate. They should not be the controlling evidence from which to locate lost corners; they may be used as corroborative and supporting evidence to give added weight to these lost corners after they have been located by other methods or means.

12.5 SUBDIVISION MACRO BOUNDARY WRONGLY MONUMENTED

Principle 5. *A subdivider who subdivides another's land cannot convey good title to land improperly monumented. After a period of time, however, a person may acquire title to an improperly monumented strip of land through unwritten methods of title transfer.*

Referring to Figure 12.1, a subdivider who inadvertently subdivides and sells another person's land may be able to solve the problem and clear title to the mislocated lots by purchasing or exchanging the land in question and making the title granted by applying the doctrine of after-acquired title, which is recognized in most states. It should be noted that in Figure 12.1, if the monumented lines are accepted, a triangular portion of land will result at the midsection line that still is in the ownership of the subdivider.

Limitations on the Principle

A subdivider who subdivides another's land may later purchase the adjoining land and, by after-acquired rights, automatically clear the title to the land subdivided. In the case of Hillside Acres (see Figure 12.1), title to the triangular strip of land on the east side could not pass because the subdivider did not have title to the land. Under certain circumstances where a person occupies land with color of title, pays taxes on the land, and so on, he or she may acquire a fee title by adverse rights. If you, as a surveyor, are hired to survey the east half of the southwest quarter of section 3, and you find the encroachment of Hillside Acres on the east half as here, you should note the encroachment on your map whether or not you considered the land lost by adverse rights. The surveyor should point out to the client what rights the adjoiner has to encroachments of long standing and also advise the client to consult an attorney. Along land grant lines, especially where the grants are several miles long, subdivisions are frequently found to overlap or not touch the true grant lines. When the cost of running out the true line is excessive or difficult, surveyors often fail to establish the line properly.

12.6 SUBDIVISION BOUNDARIES INCORRECTLY DESCRIBED

Principle 6. *A subdivider who describes the boundaries of a subdivision incorrectly but owns all the lands monumented conveys title to the land described improperly.*

A landowner petitioned the county court to lay off a town site completely within the north 40 acres of his property. The town was laid off in error, in that the south boundary extended into the south 40 acres, at his direction. Although a plat was filed depicting the town to be located properly, the court held that all sale of lots created and their conveyance was to the lots and land as monumented.³

CONFLICTING ELEMENTS IN DESCRIPTIONS

12.7 GENERAL COMMENTS

The control afforded conflicting elements *within* a subdivision, but not necessarily the boundaries of a subdivision, are expressed in the following rules. The boundaries of a subdivision, described in the map title or subdivider's title, are controlled either by the rules for metes and bounds surveys or by the rules of construction for a larger subdivision from which the new one was carved. The rules presented in the following pages are to be interpreted as applying to lots within a subdivision or to a portion of the lots within a subdivision, but not to the boundary lines of a lot abutting on a subdivision, unless specifically included.

12.8 ORIGINAL METHOD OF CREATING LOTS

Lots placed within a subdivision may be created in several ways. They may be created in an original survey with the corners monumented or they may be created by protraction, on paper, without benefit of actual survey. The retracing surveyor must know which method was used, because the procedures for retracement of the lots are different.

Principle 7. *If interior lots of a subdivision were originally created by protraction, the lots are located by measurements from the subdivision boundaries; if interior lots of a subdivision were created originally by survey with monumentation, the original survey and monuments control lot locations.*

To determine *what* must be done may require extensive field and office search—a requirement, regardless of costs.

12.9 INTENTION OF THE PARTIES

Principle 8. *The intentions of the parties to a subdivision are paramount to all other considerations.*

As one practices and testifies concerning boundary and title problems, attorneys can be expected to discuss and argue this principle in the belief that they understand what they are doing. Surveyors may be asked to testify as to what the intent of the

parties was when a description was written or a plat was prepared. That question is impossible to answer, in that only the original parties to the original deed know, with certainty, what it was they intended. This can be solved when writing the deed in the supplemental clauses. If the parties state what the intention is, there will be no question at a later date. Such statements as “It is the intent of this deed to convey the land only to the south right-of-way line and that line will be the true and correct boundary” or “It is the intent that no riparian rights be conveyed,” which detail the intent at the time of the conveyance, will solve future problems.

By most state laws, the subdivider must subdivide and survey the land before offering any one lot for sale. The intent of size and shape of all lots is thus platted and caused to be surveyed by the subdivider in accordance with current subdivision laws. The monuments set by the original surveyor to show the lines as marked and surveyed express the intent of the subdivider and become the paramount control for resurvey within a recorded subdivision. This is not true for the exterior boundary lines of a subdivision, where prior conditions, rights, surveys, or deeds may dictate otherwise.

One must be extremely cautious not to confuse intent of the parties with what the parties intended. The intent of the document is what the parties stated in the document, not what they meant to say or some unexpressed intention. In addition, the meaning and intending clause is not the parties’ *intent*. Most courts are in agreement that the meaning and intending clause is an aid to trace the title, or to supplement the parties’ previous language, not to express their intention.

12.10 FINALITY OF ORIGINAL LINES

Principle 9. *Once a lot, street, or block line within a subdivision is established by the original surveyor and the land is sold in accordance with the original plat, the lines originally marked and surveyed are unalterable except by resubdivision. Within the subdivision, boundary lines may be an exception.*

In the GLO states, this principle is fixed by federal law. The Land Act of February 11, 1805, states emphatically that the original lines and corners are unalterable by retracing surveyors. In fact, not even courts can alter or change a section line or a section corner.⁴ If land boundaries are to remain unalterable and fixed in position, no subsequent retracing surveyor should be permitted to relocate them based on more precise methods of surveying. This also applies to private subdivisions. The lines, corners, and monuments set by the creating surveyor, regardless of how imprecisely they were located, must remain fixed in place. No subsequent surveyor has the authority to “correct” any errors that are found. To do so would wreak havoc on possession, structures, and other improvements within the subdivisions. Neighborhoods that have enjoyed a long history of peace will be thrown into total disorder.

If land is to remain fixed in position and not altered by every resurvey, the principle must stand. It would indeed be folly to alter the lines of a survey and the location of the improvements thereon just because the original surveyor failed to set

his monuments in the measured position called for. The entire foundation for the stability of the land depends on this rule.

All rules of law that have been adopted for guidance in locating disputed boundary lines have been directed to the end that in so doing the steps of the surveyor who originally projected the lines on the ground may be retraced as nearly as possible. No rule that has been adopted to accomplish that end is more firmly established than that courses and distances are controlled by marked and fixed monuments.⁵

Limitations on the Principle

This principle does not apply to the lines of later surveyors or to subdivision boundary lines other than free lines. If a later surveyor establishes lines and purports them to be the original lines of the original surveyor, they can be accepted only if they are in fact the original lines. Later surveyors are not empowered to place the lines in any position other than those established by the original surveyor, and if the later surveyors establish the lines erroneously, the lines must be reestablished in their original position. Where the original surveyor of a new subdivision establishes the boundary lines of the new subdivision erroneously, and the subdivider did not own the adjoining land nor has he acquired title to it since the new subdivision was filed, the boundary lines erroneously established cannot be considered unalterable unless title by after-rights, prescription, agreement, estoppel, or other unwritten means has set in. When not in agreement with the original subdivision determining the boundaries, the boundaries of the newer subdivision may be altered because of the permanence of the lines of the older subdivision.

The boundary lines of older subdivisions established in error often become the true boundary lines by prescriptive title. If the boundaries of a subdivision have been in error and improvements have been maintained along the erroneous lines for the statutory limit, it is probably safe to assume that the adjoiner would be barred from asserting rights to the true line. This, of course, will depend on the specific facts of each situation.

12.11 CONTROL OF ORIGINAL MONUMENTS WITHIN SUBDIVISION BOUNDARIES

Principle 10. *In a lot and block description, subdivision monuments called for on the plat, or monuments set by others and known to perpetuate the position of the original monuments called for, if properly identified and undisturbed, control the position of the original lot lines.*

The retracing surveyor must thoroughly understand the difference between existing (original) and obliterated corners. The principal difference is the evidence accepted to prove them. There is no difference in their legality. In descriptions of the type “lot 2, block 3, map 2701,” nothing other than data given on the map is called for by the writings. When the map is the sole written consideration, original

monuments are paramount. Monuments, as placed by the original surveyor of a subdivision, represent the true location of the lines as run by the original surveyor, and as such have prior rights over any informative call for angles, distance, or surface. Errors in the measurements of distance, angles, or surface occur, but the location of original monuments, if undisturbed, is certain and conclusive as to the original location of the lines run by the original surveyor. Calls on plats for distance, angles, or surface are informative terms to aid in the location of monuments; the monuments mark the lines as run.

A plat is a subdivision of land into lots, streets, and alleys, marked on the earth and represented on paper. The monuments or marks placed on the ground by the surveyor in making a survey constitute the survey, and the courses and distances are only evidence of the survey. Although evidence based on courses and distances from other known points is admissible as evidence to fix a corner, where no corner is found, it is not admissible to change the location of an original corner of the survey when found.

The survey is composed of the monuments placed on the ground by the creating surveyor and the maps and plats prepared to describe that survey. Courses (bearings) and distances must be considered as evidence as to where the corners, with their monuments, are. They are “finger pointers.”⁶ These course and distances can be used as evidence from other known points to fix or to help find a corner; however, when found and identified, they cannot control a found original corner and change its position.⁷ When it comes to a difference between the found corners on the ground and the course and lines depicted on a plat, the plat is nothing more than a subdivision of land, marked on the earth and represented on paper.⁸ Numerous other court cases can be cited that recognize the control of original monuments. In Iowa, the surveyor set original monuments and a few years later set other monuments in the correct position as indicated by his field notes.⁹ An original monumenting error caused a loss of about 4 feet in Cottage Grove Avenue, a gain of 4 feet in University Avenue, and a shifting of all lots by about 4 feet. As the earliest purchasers had found the original stakes and relied on them and constructed improvements in accordance with them, the court ruled that the original stakes must control. The original surveyor had no right to alter the position of the lots after the new owners had acted with respect to the stakes set even though they were not in accordance with his plat and field notes.

When a monument is found but not identified as an original monument set by the original surveyor or as a replacement of an original monument, the control given the monument is often subordinate to distance and angles given on the original map. Having no locative value, an uncalled-for monument cannot be considered an original monument and cannot be given priority over distance or angle as given on the original map. In older subdivisions in which all the original stakes are gone as well as the records of any replacement of original stakes, the burden of proving that existing city engineers’ monuments or tie points are replacements of original stakes is impossible. If the tie points are in agreement with street and building improvements, and the improvements represent monuments built on the original lines, the tie points can be accepted.

12.12 TITLE MONUMENTS

Principle 11. *Monuments other than the original monuments or replacements of the original monuments may become title monuments by prescription, agreement, estoppel, or other means; however, these monuments cannot be used to determine original lines of the subdivision or to control such lines.*

A retracing surveyor may find several classes of title monuments. The original title may be modified or altered by unwritten legal principles, including agreement, estoppel, and others. A monument found but not located in accordance with the written deed or map may be a correct title monument, yet it may not be an indicator of the original lines of a subdivision. When a monument within a subdivision becomes a title monument by agreement, estoppel, adverse rights, judicial proceedings, or any other means, and where the monument is not an original, the monument, although binding on the adjoiners of that particular line, has no effect on the location of the original lot lines of the original surveyor; that is, it is local in character. It also cannot be used to control the replacement of other original monuments.

12.13 CONTROL OF MONUMENTS OVER PLATS

Principle 12. *Original monuments set on the ground, except where the intent is clearly otherwise, control facts given on a plat.*

The lines marked on the Earth represent the true full-scale map of the subdivision; the lines as marked on paper are a shorthand representation of what the surveyor purported to do. When there is an inconsistency between the map and the facts on the ground, the map must yield to the facts on the ground.¹⁰ When facts cannot be established on the ground—that is, the lines were never run on the ground or are lost completely—the data on the map are the best available evidence.¹¹

According to most state laws, where land is described by lot and block numbers and several parcels are created simultaneously by the filing of a map, the land must be surveyed before the sale of any one lot. In such cases, the lines as marked by the original surveyor represent the intentions of the subdivider and control resurveys.

12.14 CERTAINTY OF MONUMENT IDENTIFICATION

Principle 13. *After due allowance has been made for weathering, deterioration, and other disruptive forces, the monument and its markings should be substantially the same as the record describing the monument. In the absence of visual evidence of a monument, location of a former monument can be determined by competent witnesses who saw the monument and remembered its location.*

Time takes its toll on monuments; some alter more quickly and differently than others. From the standpoint of permanence, stability, visibility, and certainty of

identification, concrete monuments rank high, iron stakes intermediate, and wood stakes low. Pine stakes rot and disappear after a short life span; in moist locations, five years is the most that can be expected for positive identification. Redwood is well known for its resistance to decay and termites. In the Eureka Lemon Tract, many of the original 2-inch by 2-inch redwood lot corner stakes were found after a 60-year existence, and, from the excellent state of preservation, another 60 years could be expected. In the National Ranch, 3-inch by 3-inch by 4-foot scribed redwood posts, set for corner locations, were found well preserved after 82 years. In the southern United States, “lighter wood” posts were identifiable 150 years after placement by GLO surveyors.

Iron stakes in some locations are not nearly as permanent as is generally supposed. At one location, a 3/4-inch ungalvanized iron pipe was rusted away almost completely at the end of four years, whereas at another location little rusting took place in the same length of time. At the end of 25 years, 2-inch ungalvanized pipes set at Sunset Cliffs were completely rusted, whereas the redwood cores, driven in the pipes, were in good condition.

Of all devices invented by human beings, the bulldozer disrupts more stakes than any other. Monuments yield to the power of the dirt-happy dozer operator, pushing the earth around at will. In one subdivision, after all improvements were in, not a single lot stake or boundary stake was left—construction work destroyed them all. It is imperative that work in a subdivision be conducted so as to eliminate this problem.

12.15 RECORD DESCRIPTION OF MONUMENTS

Retracing surveyors have found much to complain about when attempting to identify original monuments set by the creating surveyors. The liberal use of abbreviations such as IP, IPS, and WS have made the job of retracement most difficult. With the advent of electronic data collectors, we may find once again that descriptions of monuments set at corners may be less than acceptable. Surveyors are often careless in describing monuments. The description of pipe sizes is an example. Common sizes of pipes do not measure the same as the stock designation; for example, a 1½-inch pipe measures 2 inches in outside diameter (O.D.). In reporting the size of a set pipe, many take the actual outside measurement, but others give the stock designation. If the outside diameter of a 1½-inch pipe is measured, it should be noted as 2 inches O.D., not as a 2-inch pipe. Early survey records were notoriously poor in monument descriptions, which today makes the job of retracement less certain and more difficult. Most modern laws have rectified this by requiring descriptions of all monuments set.

12.16 PRINCIPLES FOR PRESUMED CONTROL BETWEEN CONFLICTING MONUMENTS WITHIN SUBDIVISIONS

Early case law seldom referred to the term *natural monument*. Case law did refer to a *natural boundary*, but somehow, through years of legal decisions, courts have coined the term *natural monument*. Excluding possession considerations, two or

more original monuments called for in a description that are in conflict with one another are given control in the following order of presumed importance:

1. Original natural monuments [boundaries].
2. Original artificial monuments set within a subdivision.
3. Original monuments set to mark the boundary lines of a subdivision will yield to senior rights in the event of an overlap. In the event of a gap between the subdivision boundary and an adjoiner, the lines marked and surveyed will control over the call for an adjoiner.
4. Uncalled-for monuments may become controlling by common report and/or acceptance.
5. A series of boundary improvements built soon after the original stakes were set, in agreement with one another and long acquiesced to by adjoining owners, are sometimes better evidence of original survey lines than are measurements of angles and distances from other points.
6. Where two monuments, otherwise equal, are in conflict, the one in harmony with distance, angle, or area becomes controlling.
7. Judicial decisions affecting that particular subdivision. A major portion of the surveyor's work when surveying a lot within a subdivision is the relocation of monuments set by the original surveyor or the relocation of points set to perpetuate original monument locations. Land surveyors can be considered detectives who have specialized in the art of finding and relocating monuments and lines run by the original surveyors. In the absence of found monuments, other factors, such as distances given on a plat, become controlling and will be discussed under the establishment of streets and lots.

12.17 EXPLAINING PRINCIPLES

Except along the boundary lines of subdivisions, senior rights do not exist, thus accounting for the greater importance of monuments within subdivisions. Any principle given to determine the control between different types of monuments is merely a rule of construction to put in force the intentions of the parties, and any rule of construction is rebuttable and may be overcome by contrary evidence. In conveyances, natural monuments in the form of lakes, rivers, and the like are given preference because of the certainty of their identification. Artificial monuments representing the lines originally marked and surveyed are given preference over record monuments, such as the line of the adjoiner, because of the greater certainty of location. Along subdivision boundaries, artificial monuments cannot overcome senior rights unless prescriptive rights have ripened into a fee.

Control of Artificial Monuments

If artificial monuments set or used by the original surveyor represent the lines as actually run by the original surveyor, the artificial monuments are presumed superior

to record monuments. Thus, in Figure 12.2, the original surveyor assumed in error that the iron pin found at the northwest corner of Olive and A streets was the true corner and started a subdivision from the erroneous point. To relocate a lot within this subdivision, the surveyor should use the pin as his or her starting point and overlook the call for the record monuments (street line). The street as originally staked is 76 feet wide and is unalterable except by conveyancing. In a New Jersey case, very similar in nature, the opposite was held. The court reasoned that the unmarked street was a natural monument (*sic*) that should control the artificial monument.

Meander Lines

Meander lines are quite unusual, in that they were never intended to be considered as boundary lines. Initially, they were to represent the natural curvatures of the water body or other feature that was being surveyed. The primary purpose of a meander line was to permit the calculation of area to the proximity of the water body or other boundary. Artificial monuments set by a surveyor to meander a natural monument, such as a lake, river, or ocean, must yield to the more certain monument—that is, the waterline. Meander lines are run for the purpose of plotting the natural monument being meandered and to permit the determination of area. Meander lines should not be considered as marking the boundaries or limits of the natural monument or of lots adjacent to it. “Where a meander line of land included in a patent is run to locate a shoreline approximately and to afford the means for calculating the area of the land granted, the shoreline constitutes the real boundary.”¹²

This presumption may not be applicable in true metes and bounds descriptions.

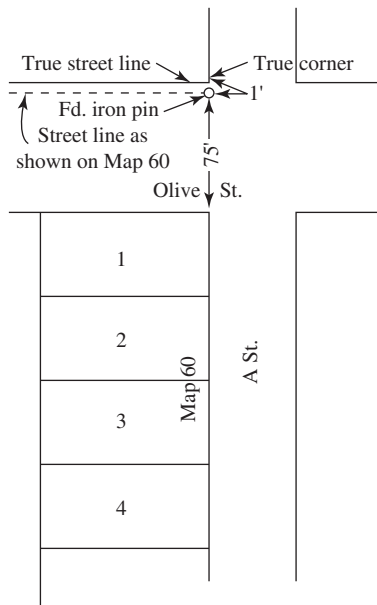


Figure 12.2 Case in which an artificial monument is superior to a record monument.

Uncalled-for Monuments

Monuments that are not called for are uncertain; to be controlling they must be proved. If they are found in the proper position for angle and distance, they may automatically become the correct corner markers. If they are not in the measured position, they cannot be accepted without substantial evidence showing that they are in the position of the original stakes or that they have been accepted by common report.

Unfortunately, in many older subdivision surveys a statement explaining what monuments were set was not required by law; yet we know that the lands were surveyed and points were set. A statement on a map "Surveyed by Wheeler in 1880" is conclusive proof that the land was surveyed, and it must be presumed that monuments were set.¹³ Notes taken at the time of early surveys amounted to private property, and as such have been lost as the years advanced. In some instances, a thorough resurvey reveals consistent types of corner material set by the original surveyor. In Rancho Mission, 4-inch by 4-inch redwood posts about 3½ feet long, set 1 foot in the Earth and stenciled with black paint indicating the corner numbers shown on the original map, were found as corner monuments. In a superior court case involving lot line location in Wadsworth's Olive Grove, 2½-inch by 2½-inch by 18-inch redwood stakes were accepted as the original stakes even though not mentioned on the original map. In both of these examples, the principle that a monument should be called for before it is controlling is defeated because of the preponderance of evidence proving what was actually set, even though not described on the maps.

This principle is sustained by the court in its findings:

In an action to quiet title involving a disputed boundary between lots, referring to a map uncertainly locating the line, and no field-notes of the survey for the map are indicated, it is proper for the court to receive the best evidence obtainable to determine the location as a matter of fact, and in so doing, was important to inquire as to the location of such stakes and monuments as were commonly recognized, accepted and used in lieu of lost or destroyed original monuments, and in the absence of more certain evidence, these stakes and monuments would be sufficient to support a finding as to the location of the boundary.¹⁴

In this case, the court accepted parol descriptions of the old monument found but not described on the original map. One surveyor testified: "In making surveys here in the city [Los Angeles] the last 13 years I have had occasion to retrace a great many of those Hancock corners. There were a few of them left and that was one of the last to go." Although this is considered to be hearsay, it is *considered* an exception to the hearsay rule.

Common Report

By reputation, certain markers are commonly accepted by surveyors and others as being correct, even though they cannot positively be proved to be correct. As time progresses and the records of replacement of original property monuments are lost,

surveyors become more and more dependent on the acceptance of monuments whose history is lost in antiquity. If people accept a monument, and especially if numerous surveyors accept a monument, the monument is said to be the true marker by *common report*, or what contemporary courts call “repose.” Monuments that can be proved to be neither correct nor incorrect and are commonly accepted as being correct may be accepted by common report, but a monument that can positively be proved to be incorrect, although accepted by many as being correct, cannot be accepted by common report. Mere measurements from distant objects does not disprove a monument. But finding an original monument not in agreement with a 1/2-inch pin accepted by common report certainly refutes the pin. For a monument to be accepted by common report, the following factors should be present: (1) The monument is commonly accepted and has the reputation of being correct, (2) the monument is in a position that represents a spot or place marked by an original surveyor or a spot that could have been marked by an original surveyor, and (3) the monument cannot be disproved. Several courts have also referred to this doctrine as the *doctrine of repose*.

Improvements as Monuments

A resurvey made after the monuments of the original survey have disappeared is for the purpose of determining where the monuments were, not where they ought to have been.¹⁵ If the original monuments of an original survey are lost and a series of old fences, old buildings, or other ancient boundary indicators are in agreement with one another, it is usually presumed that the improvements were built on the original lines of the original surveyor and stand as monuments representing the original lines. Such improvements are considered as being the best remaining evidence of the original survey. This rule has more force when it can be shown that the improvements were made soon after the original stakes were set. Many tie points are set in accordance with this principle. When there is uncertainty about city tie points that represent the position of the original monuments and several surveyors are uncertain about the true location of the original monuments, the courts will frequently settle on a fence of long standing as being the better evidence of a boundary line. Acceptance of improvements is a principle that should not be applied if there is better evidence of the original positions of the monuments, and such evidence as fences should have no greater weight than any other element of evidence. Associate Justice Cooley, in 1878, stated in his opinion:

This litigation grows out of a new survey recently made by the City Engineer. According to this survey the practical location of the whole plat is wrong, and all the lines should be moved four or five feet east. . . . When an officer proposes thus dogmatically to unsettle the landmarks of the whole community, it becomes of highest importance to know what was the basis of his opinion. The records in this case fail to give an explanation. Nothing is better understood than that few of our early plats will stand the test of a careful and accurate survey without disclosing errors. This is true of the government surveys as of any others, and if all the lines were now subject to

correction on new surveys, the confusion of lines and titles that would follow would cause consternation in many communities. Indeed the mischiefs that must follow would be incalculable, and the visitation of the surveyor might well be set down as a great public calamity.

But no law can sanction this course. The surveyor has mistaken entirely the point to which his attention should have been directed. The question is not how an entirely accurate survey would locate these lots, but how the original stakes located them. No rule in real estate law is more inflexible than that monuments control course and distance . . . the city surveyor should therefore, have directed his attention to the ascertainment of the actual location of the original landmarks, and if those were discovered, they must govern. If they are no longer discoverable, the question is where they were located; and upon that question the best possible evidence is usually to be found in the practical location of the lines, made at the time when the original monuments were presumably in existence and probably well known. As between old boundary fences, and any survey made after the monuments have disappeared, the fences are by far the better evidence of what the lines of a lot actually are.¹⁶

Also, Justice Hand noted: “In case of a disputed boundary line in a town, city or village, where the monuments from which the town, city or village was platted are lost or destroyed, the courts ought not to disturb boundary lines between lot owners which have been acquiesced in for years and upon which the lot owners have erected improvements.”¹⁷

12.18 INTRODUCTION TO PROPORTIONING

Principle 14. *Within a subdivision, distance, direction, and area are presumed subordinate to the intent of the subdivider; the lines are as marked and surveyed, and original monuments control. Where excess or deficiency is found, it is distributed in proportion to linear measurements between original found monuments.*

Few landowners or courts fully understand proportioning of excess or deficiency. The one basic principle underlying all proportioning is that proportioning can occur only between found original corners of lots. Although in many states direction is considered more certain than distance, excess or deficiency discovered in subdivisions is distributed in proportion to linear measurements. This is probably because frontage on a street is considered as paramount. Even though most surveyors know that distance is low in priority and may be a mere estimate, it is often the most important item in the layperson’s mind. Usually, explanations to clients must be made.

In summary, the primary objective of retracing a boundary is to locate and prove, by the totality of best evidence available, where the original corners were placed and the remains of the monuments set to identify these corners. It does not matter whether the original corner monument was destroyed and the corner itself is difficult to position. The surveyor is permitted to use originality and the courts will permit flexibility of recovering and interpreting evidence to prove the original positions.

ESTABLISHMENT OF STREETS

12.19 GENERAL COMMENTS

Boundaries of streets are established by the following methods listed in their usual order of importance: (1) by natural monuments, (2) by artificial monuments and lines actually run at the time of making the plat, (3) by improvements, (4) by the line of nearby streets where called for, (5) by the data given on the plat, and (6) as a last resort, by proportional measure. As the lines of streets and blocks are usually one and the same thing, the establishment of street lines normally establishes block corners.

12.20 ESTABLISHMENT OF STREETS BY NATURAL MONUMENTS

Principle 15. *Natural monuments, where called for, are presumed to control street lines.*

Figure 12.3 shows a portion of Bird Rock subdivision with Ocean Boulevard existing between Blocks 20 and 21 and the ocean. This defines the boundary of Ocean Boulevard as being riparian and extending to the ocean regardless of the distance given or scaled on the original map. To define it otherwise in this case would defeat the intentions of the subdivider, who has shown clearly by the plat that the intention was for the street to extend all the way to the ocean.

12.21 ESTABLISHMENT OF STREETS AND ALLEYS BY ARTIFICIAL MONUMENTS AND LINES ACTUALLY RUN AT THE TIME OF MAKING THE PLAT

Principle 16. *After natural monuments, artificial monuments that represent the actual lines run by the original surveyor at the time of making the plat are presumed*

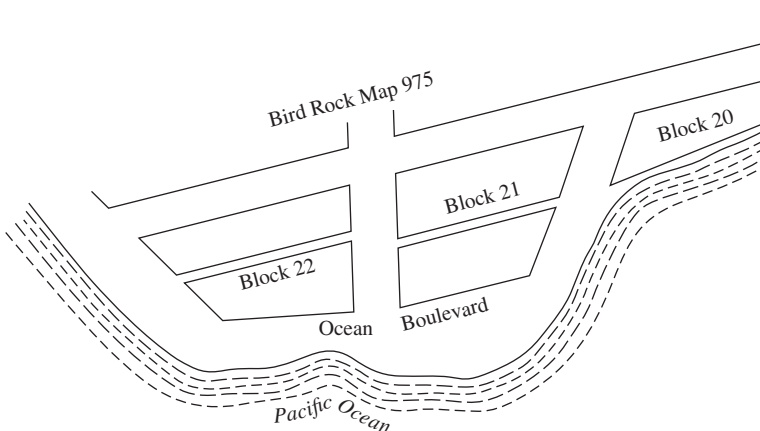


Figure 12.3 The boundary of Ocean Boulevard is riparian and extends to the ocean.

to control street lines irrespective of whether the courses, distances, and street improvement agree with the plat.

Identified original monuments set by the original surveyor and found undisturbed will control the street line as shown in Figure 12.4 Street A was found to measure 50 feet between monuments instead of 40 feet as indicated on the original map. The adjoining lots cannot each receive 5 feet extra by narrowing the street; the artificial monuments found definitely establish the street line. Similarly, if the record measurement is greater than the actual width indicated by monuments, the streets cannot be widened at the expense of the adjoining lots. In this case, the subdivider's intentions are clearly shown by the markers set at the time of the subdivision. The fact that the mapmaker erroneously noted 40 feet instead of 50 feet does not alter the facts on the ground.

In the city of Decatur, Illinois, an alley was located by the city crew and again by the county surveyor; neither found an original stake, but both agreed that by measurement the alley should be farther north and that this location would include the enclosure and parts of the buildings of the defendant. The opinion of the court was as follows:

A. T. Risley, a former county surveyor, testified that when surveying he found a number of stakes on the line of the alley as now located, and recognized some of the original corners in Block 2; that the alley in block 2 was on a line with the alley in block 1, and that the line is substantially the same now that it was then. . . . That the last time he [Niedermeyer] re-built the fence, nine or ten years ago, he found one of the original pegs in the ground. All the witnesses testified that the fence had been on the same line ever since it was first put up; that no change had been made in the location of the alley since it was first fenced on both sides. . . . This evidence is uncontradicted. . . . As has been repeatedly held by this court, the true boundary lines of a city lot are where they

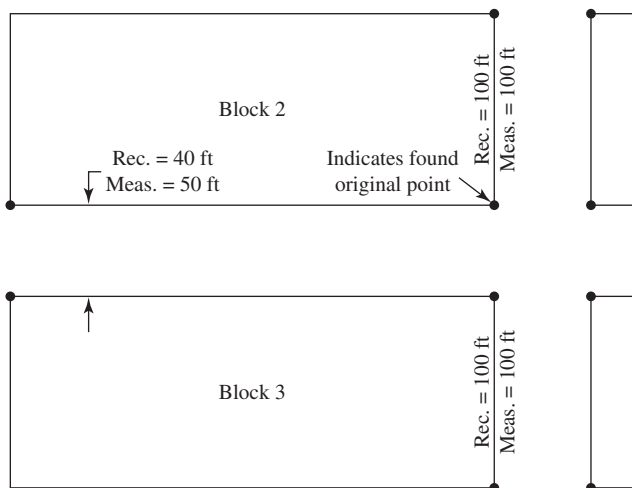


Figure 12.4 Streets are not of record width when original monuments indicate otherwise.

are actually marked by the monuments placed by the surveyor to indicate where they are to be found, and the most satisfactory evidence of the place where the lines were located is afforded by the original stakes.¹⁸

Owners of the land south of Cheltenham Avenue in St. Louis could not claim 20 feet of the surplus in the subdivision merely because they were entitled to a prorated share of the entire subdivision.¹⁹ Stones placed in Cheltenham Avenue by the original surveyor fixed the position of the street; all surplus belongs to the land north of the street.

Street improvements not built in accordance with the original stakes set by the original surveyor cannot change the street alignment; the original control is conclusive and cannot be altered. This does not preclude a city or other governing agency from claiming the streets by prescription; it merely means that a street claimed by prescription, if properly described by written title, must be described in accordance with the original lines as marked and surveyed by the original surveyor if such lines can be proved. The city must meet the same burden as that met by a private citizen when proving title by prescription.

12.22 ESTABLISHMENT OF STREETS BY IMPROVEMENTS

Principle 17. *In the absence of natural monuments or evidence of lines actually run by the original surveyor, improvements, such as curbs and paving, which were installed in accordance with the original survey monuments, are presumed controlling.*

The duty of the surveyor is to relocate lines as they were originally run by the first surveyor; curbs located properly when the original stakes were available may be the best evidence as to where the original lines were run, and as such are controlling. The court observed:

In determining the line of a street, measurements upon such street are of more value than measurements taken elsewhere; and if they or the places where they were cannot be located, the boundaries of the street as actually opened and used should be ascertained; and if such location has been generally acquiesced in by the public, by lot owners and by the municipality, in the *absence of more certain evidence*, it will be conclusive.²⁰

When the reason for the principle ceases—that is, the improvements were not built in accordance with the original stakes—so does the principle. An engineer's failure to place the construction stakes in accordance with the original stakes does not allow the street to be moved in position to cover a construction blunder. When a street is long used by the public and the location of the improvements are not in accordance with the original monuments, the public may acquire a right, by prescription, to that portion improved. This *does not* alter the original street lines; they are unchangeable. Where by reasonable analysis it can be shown that the original street lines and the present improvements are not in agreement, the original block corners may not be in the side lines of the streets as obtained by prescription.

Home Avenue was constructed and staked by locating the street from Swan's Addition, which was a resubdivision of a portion of Wadsworth's Olive Grove.

When attempting to locate Home Avenue as shown on Wadsworth’s Olive Grove, the surveyor measured the record distance as shown on said map and set the side line markers for Swan’s Addition. He failed to note that the measurements extended to the centerline of the street and thus located the street incorrectly by one-half its width. The Superior Court ruled that the starting of a survey from the side line of Home Avenue as established by Swan’s map was improper for the establishment of a lot in Wadsworth’s Olive Grove. Home Avenue had been built and improved for more than 20 years in its incorrect position; yet the improvements alone, unless built in accordance with the original stakes, are insufficient to control original lines. If there had been a lack of evidence to show where Home Avenue should have been with respect to Wadsworth’s Olive Grove, the court might have ruled differently.

12.23 ESTABLISHMENT OF STREETS BY THE LINE OF A NEARBY STREET

Principle 18. In the absence of artificial monuments and evidence of the lines as marked and surveyed, where a street is plotted as being the continuation of a nearby street or beginning at a nearby street, the line of the nearby street is presumed to control.

The call for the line of a street is a call for a record monument and, as such, is presumed to control over distance and angle. Figure 12.5 shows part of the Frary Heights subdivision, where Palm Street is shown as a continuation of the street of

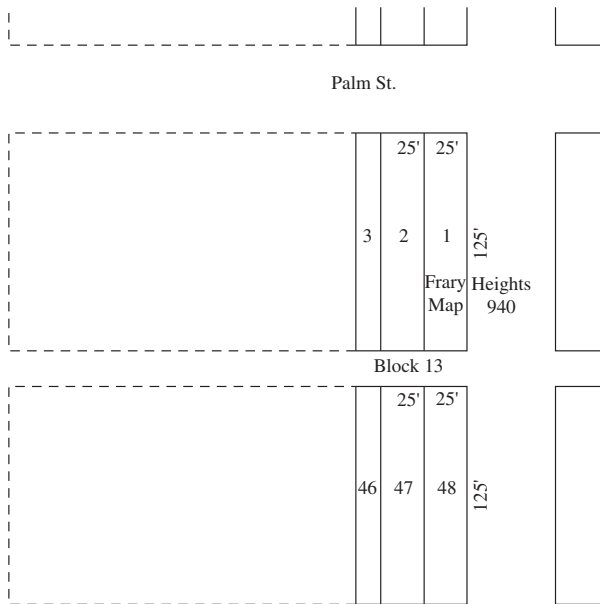


Figure 12.5 Where a street is plotted as the continuation of a nearby street, the line of the nearby street controls.

the adjoining subdivision. In the absence of finding original monuments within the subdivision on Frary Heights, the continuation of the line of Palm Street as found in the adjoining subdivision is proper. This principle is defeated where found original monuments indicate that the street line is not a continuation of the adjoining street.

12.24 ESTABLISHMENT OF STREETS BY PLAT

Principle 19. *In the absence of evidence covered by the foregoing principles, the exact width of the street as given on the plat and the distances and angles are presumed to govern street location. Occasionally, a measurement index is applied.*

In the absence of monuments, streets are given the width called for on the plat, regardless of excess or deficiency that may exist within a subdivision. In Figure 12.6, two original 2-inch by 2-inch hubs found two blocks apart measured 2 feet in excess of the record distance. The proper procedure is to give Blocks 1 and 2 exactly half of the surplus (1 foot each) and give the street exactly 50 feet, as called for by the map. The rights of the public to a street are thus protected by the courts so that deficiency or excess cannot exist within a street except where the original monuments set by the original surveyor indicate otherwise. In a few jurisdictions, a measurement index is applied.

Measurement Index

Not all surveyors and courts are in agreement with the theory that excess or deficiency is not to be prorated in the width of the street. If the original surveyor, establishing monuments within a subdivision, used a long or short tape, the streets as well as the blocks would be equally long or short. The uniform excess or deficiency is referred to as the *measurement index*. In Long Island City, New York, the chain used to tie a point was 100.08 feet instead of 100.00; the custom is to prorate 0.04 foot in a 50-foot street. In San Diego, California, Horton's Subdivision has a fairly uniform surplus of $\frac{1}{2}$ foot to a 300-foot block, yet streets are always staked at exactly 80.00-foot widths and all surplus is given to the blocks.

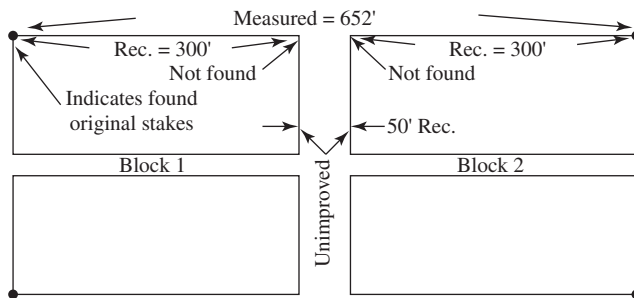


Figure 12.6 In the absence of monuments, streets are given the width called for in the plat.

The measurement index may also be considered in conducting a GLO retracement. If a surveyor is not able to recover two original or obliterated corners, if in remeasuring between those two corners the surveyor determines a difference between the original measurement he or she obtained and the original measurement, this factor may be considered in looking for additional evidence not possible in surveying lost corners.

A measurement index can be applied where there is a uniform overage or shortage in many blocks within the same subdivision, but it cannot be applied where there are erratic measurements that do not indicate a uniform error. In a subdivision where the blocks are supposed to be 300 feet each and the actual measurements indicate 300.25, 300.35, and 300.30 feet, it might be assumed that the original tape was in error. If the measurements were 299.75, 300.25, and 300.00 feet, no measurement index exists.

Only one court case was found in which excess and deficiency were distributed within a public way, wherein a 43.8-foot shortage was divided between two lots and an alley.²¹ The weight of reason is against such a procedure. Before the establishment of any subdivision, street widths that are acceptable to the public are determined by the governing agency, and the subdivision is accepted by the public agency on the condition that the streets are of a certain definite width. The size of lots is determined by the whims of the subdivider, the desires of the purchasers, and the minimum area requirements of the planning agency. The tendency of all subdividers is to make the streets of minimum width so as to have a maximum amount of land for lots. The streets are definite and fixed, whereas the lots are variable. This area is particularly difficult, in that streets may be platted but not located on the ground. In doing their research in a retracement, surveyors will often overlook the streets that were not built.

12.25 ESTABLISHMENT OF STREETS WHERE WIDTH IS NOT GIVEN

Principle 20. *If the street width is not given on the plat, the width as scaled on the map will govern as a last resort.*

The complete absence of any other acceptable evidence on the ground makes it necessary to use scaling as the only means available to determine street width. Scaling should be considered as the very last resort.

12.26 ESTABLISHMENT OF STREETS BY CITY ENGINEERS' MONUMENTS

Principle 21. *Offset monuments set by the city engineer or other officials to perpetuate the position of the original monuments of the original surveyor control street lines. Offset monuments not based on adjacent original monuments are afforded control only in proportion to the accuracy with which they were set in accordance*

with the foregoing rules. City engineers' monuments long acquiesced to are presumed to be correct; the contrary must be proved.

Simply because a city engineer or the engineer's assistants set a monument does not prove the monument to be correct. Where a monument is set by measurements of angle and distance from other known original monuments, the monument set cannot be considered as controlling except where it is in agreement with the data given on the map. This is well illustrated in the case in which the original town of Sacramento was surveyed by Sutter in 1848 or 1849. In 1878, the city engineer established the street lines by starting at points many blocks apart and prorating in the intervening streets. Evidence showed that the Sutter survey and the stones set in the 1878 city survey were not in agreement. A surveyor in 1911 established a lot with respect to the 1878 stones and overlapped a fence. The court observed:

In this action in ejectment and for damages for the unlawful detention of land, which involved the location of the boundary line between two lots in a city of Sacramento, upon the dividing line of which a fence has existed for probably 40 years, or more, it is held that in view of the meager character of the evidence of the real boundary line as fixed by the original survey of the city [Sutter survey] and in accordance with which the deeds of the parties were made, and of the disputed strip in the enclosure of the defendant, the court was justified in finding that the plaintiffs had failed to establish any title to the property in controversy.²²

Here, the court accepted a fence as better evidence than uncertain monuments. Note that, although the stones set in the later survey were not acceptable for the location of lot lines within a block of the subdivision, they might be acceptable for determination of the street lines. Streets can be acquired by usage of the public even though not based on an original survey.

In a case in which the city had passed an ordinance making all surveys not conforming to the new points established by the city engineer null and void, the court observed:

An ordinance of a city providing that a certain monument shall from that date of the ordinance be the initial point of the town survey and of all locations of lots and streets, and that all surveys made thereafter, that shall deviate from such initial point shall be null and void, is void, and not admissible in evidence to show enlarged rights against the city. The city council cannot change the location of streets by such a resolution, or affect the rights of land owners under grants previously made, nor can it lay down rules of evidence by which the courts are to determine the location of the points or lines of the survey.²³

In this case, a landowner was attempting to move his fence some 16 feet to include a portion of the street that was long recognized by improvements. The shift in the lot position was brought about by using the point of beginning, as mentioned previously.

Where a deed from a city bounded the land granted by a street which had been previously located and surveyed by a city surveyor, and the grantee had for many years held

possession of and fenced the lot nearly according to such location and survey, and the street was in use by the public accordingly, the fact that a new survey of the street is afterwards made, changing its line so as to exclude therefrom a strip of land adjoining the lot granted, cannot entitle the grantee to remove his fence to the line of the street as fixed by the new survey, so as to include such strip in his lot, but he is restricted to the lot as bounded by the line of the street as surveyed *originally*.

In determining the boundaries of city lots the line as originally located on the surface must govern; and a line shown by monuments as platted by the city authorities and acquiesced in for many years must control course and distance, and cannot be overturned by measurements alone.²⁴

In summarizing the three court cases cited here, it can be said that city engineers' monuments are not controlling where they were not set correctly with reference to the original stakes. When, however, city engineers' monuments have long been acquiesced in and used by surveyors and the public, they will be *presumed to be correct* and may be *prima facie* evidence except when the contrary can be shown.

ESTABLISHMENT OF LOTS WITHIN SUBDIVISIONS

12.27 EFFECT OF MATHEMATICAL ERROR

Principle 22. *Excess or deficiency existing in a straight line between fixed monuments within a subdivision is distributed among all the lots along the line in proportion to their record measurements.*

In Block 9 of Mission Beach (Figure 12.7), the length of the north line of lot D is indicated as being 66.58 feet, but the scaled distance is 96 feet, more or less.

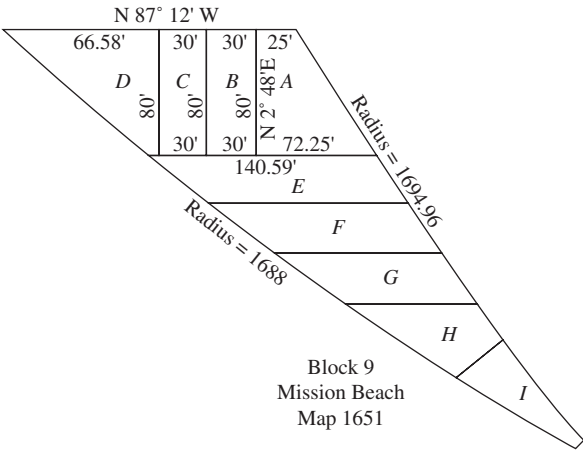


Figure 12.7 Excess or deficiency within a block caused by mathematical error is given to the lot in which the error occurs.

Mathematically, the lot will not close by 30 feet, more or less. If each lot were to receive a portion of the surplus, the lot corner common to *D* and *C* would shift some 24 feet west of that pictured and the intent that lot lines are about 90° to the north line and parallel with each other is defeated. The proper method is to give lots *A*, *B*, and *C* their record distance, which will automatically correct the mathematical error of lot *D* by establishing the length of the north line of lot *D* at about 96 feet. This refers to the principle that an “an error should be placed where the error occurs, if it can be so ascertained.”

12.28 EXCESS OR DEFICIENCY

Excess or deficiency occurs whenever a recent precise measurement between original monument positions differs from the accurate measurement originally reported between the same monument positions. Excess or deficiency often implies that the original surveyor or scrivener of the conveyance made an error or blunder, or relied on erroneous measurements. Whatever the cause, the excess or deficiency belongs to someone or several people; the problem is, who is entitled to it? As an error may have been made, the courts are reluctant to accept the principle that an error should be divided among several people. As a last resort, excess or deficiency is prorated. For example, Mr. Stringfellow dies and leaves all his land to his two sons—the west 1320 feet to the older and the east 1320 feet to the younger. After the land is measured, it is discovered that there are 2710 feet instead of 2640 feet as assumed. A whole was inherited by the two; each received equal quantities of land; each is entitled to half of the excess.

Mr. Burne dies and leaves the west 1320 feet to his older son and the remainder to his younger son. In this case, according to the written record, no matter how much excess or deficiency exists compared to the measurements reported originally, the younger receives a remainder and the older receives 1320 feet.

Mr. Brown sells the west half of his lot and retains the east half. Two equal rights are created simultaneously and each receives half regardless of whether the area or measurements differ from the record. Each party has proportional rights.

Within a block, 10 equally sized lots are protracted (not monumented) and the surveyor indicates that he set only the block corners. Each lot is entitled to a proportionate share of the entire block and excess or deficiency is divided among the lots.

Accretion forms in front of several properties and belongs to all the properties. The excess is distributed among the properties by some proportionate means. There is no hard-and-fast rule as to what is correct.

12.29 PRORATION: A RULE OF LAST RESORT

Before prorating an excess or a deficiency, an investigation of the possible causes of the excess or deficiency should always be made. In a block shown to be 240 feet

wide, a deficiency of 15 feet was found.²⁵ The lower court found that the eastern lots should get full measure because of acquiescence, but this was rejected by the upper court. In ordering a new trial, the Supreme Court observed that the shortage probably did belong to the western lots (for a different reason) and suggested that (1) the eastern lots in question, although included in the subdivision, were sold by metes and bounds prior to the filing of the subdivision and thus had senior rights, and (2) the street west of the western lots was on the centerline of a section, and, at the time the city improved the street, it probably included part of the western lots. In other words, the upper court ordered a new trial for the purpose of discovering where the shortage occurred. Proration was not to be applied if the cause of the shortage could be discovered.

12.30 EXCESS OR DEFICIENCY CONFINED TO A BLOCK

Principle 23. *Excess or deficiency occurring within a block should not be prorated among other blocks.*

Excess or deficiency in the land platted into lots, blocks, and streets along with an absence of original markers does not always indicate that intervening streets should be located by proration. If possible, each block should be treated as distinct and the shortage or surplusage therein apportioned among the lot owners.²⁶ Proration is a principle of last resort, and once a street is established by one of the means previously discussed, the street lines are unchangeable. Just because Block 1 in Figure 12.8 is 1 foot short and Block 2 is 1 foot long does not mean that the location of B Street can be moved 1 foot east in order to give Blocks 1 and 2 each exactly the record measure. All the shortage existing in Block 1 is prorated among the lots of Block 1, irrespective of the surplus in an adjoining block. Only if B Street were not established by improvements, possession, or found points could Blocks 1 and 2 each receive an equal frontage between streets. Under this principle, the surveyor is prohibited from going beyond the limits of the specific block. This is a basic principle of retracements.

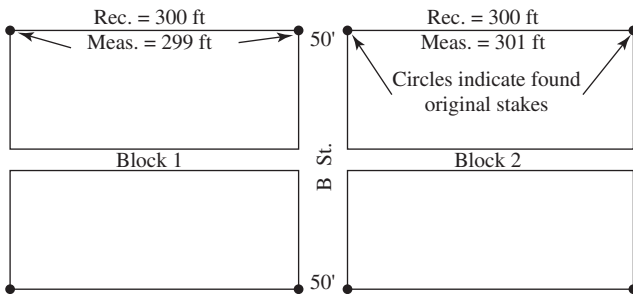


Figure 12.8 The shortage in Block 1 is prorated among the lots of Block 1.

12.31 EXCESS OR DEFICIENCY DISTRIBUTION WITHIN BLOCKS

When an excess or deficiency is found in measuring a line, it must be assumed that the error of measurement occurred not in any one part of the line, but in all of the line. The original surveyor probably had a chain that was either too long or too short, and the error would occur in all parts of the line. Where a subdivision shows 10 lots of equal frontage, it is logical to assume that any small excess or deficiency should be distributed among all lots equally and not given to the end lot.

Excess or deficiency distribution is applicable where several parcels were created simultaneously by proceedings in partition, by protraction, or by subdivision. Proration of surplusage and shortage cannot apply to metes and bounds descriptions where a senior grant exists, to blocks with undimensioned end lots, to errors the location of which can be established, to lines where monuments fix the location of the error, or to lines where points are fixed by improvements made in accordance with the original stakes.

Where land is plotted into lots, blocks and streets, and surplus exists, the surplus is to be divided among the lots.²⁷

Where in platting a village it turns out that by mistake the blocks are not so long as the plat represents, the deficiency must be apportioned between the lots of the block according to their apparent size as shown by the map.²⁸

There are many more examples in each state or jurisdiction. To fully understand each local application, the surveyor must research each specific area where retracements are to be made.

12.32 SINGLE PROPORTIONATE MEASURE

Principle 24. *Excess or deficiency existing in a straight line between fixed monuments within a subdivision is distributed among all the lots along the line in proportion to their record measurements.*

In block 1 of Hambleton, Weston & Davis's subdivision in Chicago, Illinois, a shortage of 2.68 feet occurred. By record measurement from one end of the block, a brick wall of a building was on its own lot (the north half of Lot 5 and the south half of Lot 6), but if the 2.68 feet was divided by proportion among the lots, the building encroached on the adjoiner. Assuming that proration applied, the adjoiner used the brick wall to support several girders of his new building. In approving the action, the court observed:

In the absence of any agreement or question of title by adverse possession, where a block has been platted into lots and the lots sold, a shortage in the block will be prorated among the several lots. Complainant, therefore, when it received its deed, the property being described as the north half of lot 5 and the south half of lot 6, and not

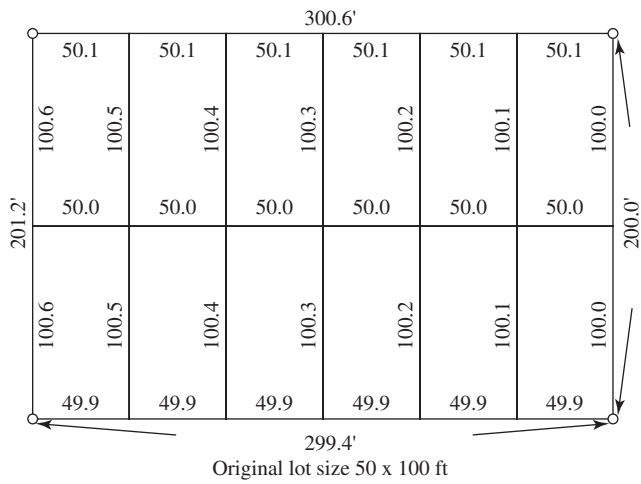


Figure 12.9 Errors of measurement are divided proportionately between lots.

by metes and bounds, did not get paper title to 509 frontage, but only received paper title to the north half of lot 5 and the south half of lot 6 after the same had been prorated by reason of the deficiency.²⁹

This principle should apply to the distribution of small errors and small differences between present measurements and original measurements. When gross errors occur, a different rule often, but not always, applies. When the block is abutting on a subdivision line, extreme caution should be exercised in locating lots by proration, since the principle frequently does not apply.

Figure 12.9 shows the application of proration to a block where the block corners are fixed by monuments. Angles at the block corners must be measured to show the angles for any one lot, but the angles have no influence on the distribution of excess or deficiency, for they are based entirely on linear measurements.

12.33 SINGLE PROPORTIONATE MEASURE ON CURVES

Principle 25. *Excess or deficiency existing between fixed monuments on a curve is distributed among the lots along the curve in proportion to their record linear measure.*

In Figure 12.10, the length of the curve along Le Roy Street was found by actual measurement to be 8.40 feet short of the original measurements. The beginning and end of the curve were fixed by street improvements and by monuments set by the city engineer; no trace of the original markers could be found. Because the beginning and end of the curve were fixed by presumption and could not be altered, Lots 4, 5, and 6 must take all the deficiency. The fixed monuments preclude Lot 7 from being short in measurement.

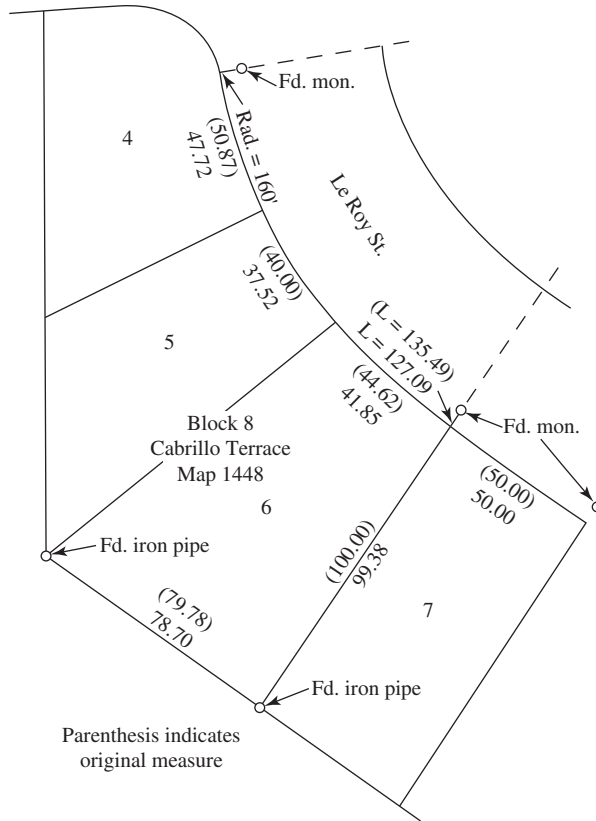


Figure 12.10 Proportionate distribution of errors along a curve.

Proration along a curve is controlled by the same rule that governs single proportionate measure; that is, each segment of the curve has the same proportionate length of the total curve as is shown on the original map. The new length of Lot 6 is calculated as follows:

$$127.09 \times 44.62 \div 135.49 \times = 41.85 \text{ feet}$$

The lot frontage then becomes 2.77 feet short of the record measurement. Frequently, the city engineer or surveyor establishes a curve that is not based on the original curve points because the curve data given on the map are out of agreement with the stakes set on the ground or because the stakes as set are not a true curve. Any survey locating lots from a curve alignment not in agreement with the original stakes has made use of a poor foundation for the survey. Surveyors should approach this area with caution.

Corners placed on curves are difficult, if not impossible to measure their position if they become destroyed. The elements of curves can be described by numbers, but

one cannot survey along curved surfaces. The elements of curves can be surveyed—the radius, central angle, chord lengths—but there is no known method of measuring a curved line that identifies the curve itself.

**12.34 DISTRIBUTION OF EXCESS AND DEFICIENCY BEYOND
A MONUMENT**

***Principle 26.** Excess or deficiency cannot be distributed beyond any undisturbed original monument.*

Undisturbed original monuments are fixed in position and cannot be overridden by proration even when the monuments are found in the middle of the block. Using the same block as that shown in Figure 12.9, the distribution of excess and deficiency is modified by the found monuments shown in Figure 12.11. Judge Marshall, in 1897, commenting on the establishment of lot corners from known corners found within a block as compared to remote block corners, observed:

The *unvarying rule* to be followed in such case is to start at the nearest known point on one side of the lost corner, on the line on which it was originally established; to then measure to the nearest known corner on the other side of the same line; then if the length of the line is in excess of that called for by the original survey, to divide it between the tracts connecting such two known points, in proportion to the lengths of the boundaries of such tracts on such line, as given in the survey. . . . The method always followed in reestablishing corners is to measure the line connecting the nearest known corners, on the same line, on either side of the lost corner, and then divide the excess, if any be found, as before stated.³⁰ (Emphasis added.)

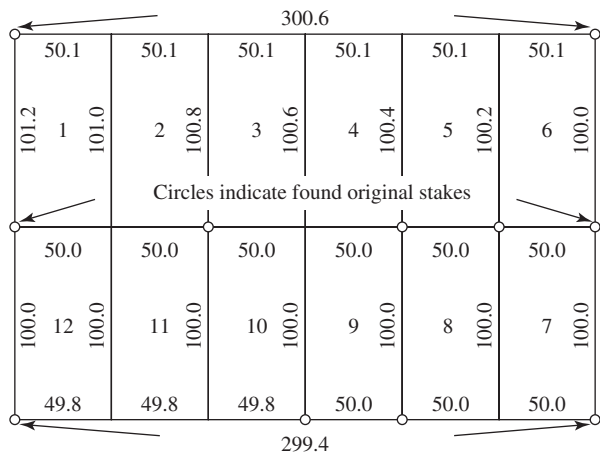


Figure 12.11 Errors of measurement are distributed proportionately between found original monuments.

Although proportioning cannot extend beyond an original monument, the original monument can be used to determine the direction of the line it controls. This will permit an alternate means of locating lost corners.

12.35 ESTABLISHMENT OF LOTS WHERE THE END LOT MEASUREMENT IS NOT GIVEN

Principle 27. When the end lot measurement is not given, all the excess or deficiency is presumed to be given to the end lot.

When the original subdivider failed to give a dimension to the last lot in the block, it can be assumed that the subdivider intended to place all excess or deficiency in the last lot. The fact that the last lot scales differently from the amount of land on the ground does not alter the principle. This situation frequently occurs along the boundary of a subdivision where the subdivider was uncertain as to his true boundary line. Figure 12.12 shows the condition in Block 13 with the general note that all regular lots are 100 by 100 feet. Because Lots 3 and 4 are true remnants of lots, all excess or deficiency is given to Lots 3 and 4 regardless of the scaled distance. Lot 3, which scales 100 feet, will receive 150 feet. In New Jersey, a map showed 50 lots: 48 were regular with a width of 25 feet each, 2 were irregular and undimensioned; the court ruled that the regular lots would receive 25 feet each and the undimensioned lots would receive what was left over even though they did not get the amount of land that the map appeared to give them.

12.36 REMNANT PRINCIPLE

Principle 28. Where excess or deficiency is given to an irregularly shaped lot at the end of a block, the method is called the remnant rule. Few jurisdictions accept it.

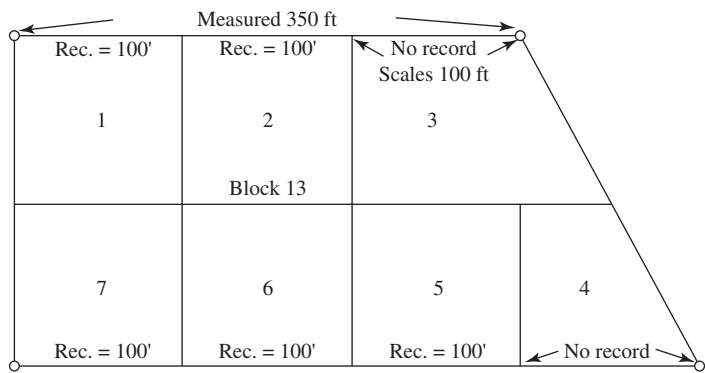


Figure 12.12 Remnant lots without dimensions receive the remainder.

Fairview Avenue, the end of the plat, is insufficient to supply the number of lots given on the plat with the dimensions stated. So that beyond question there was a mistake in the preparation of the plat, or in the original survey, and the purpose of the action is to locate and correct it. The whole controversy in the court below narrowed down to the question whether Lot 22 was or was not erroneously designated on the plat as a seventy-five foot lot. The trial court in effect found that this lot was entitled to a frontage of 75.38 feet; and, if that conclusion be sustained by the evidence and the rules of law controlling the question, the case is at an end.

We are of the opinion that the learned trial court erred in the conclusion stated. The evidence is insufficient to sustain it. The court relied, in reaching its conclusion, chiefly upon the testimony of witness Armstrong, county surveyor of Ramsey County. This witness, though apparently fair and candid, seems to have fallen into the initial error of adopting as correct the notation on the plat that Lot 22 had a frontage of 75.38 feet on University Avenue.

But conceding for present purpose the correctness of the conclusion reached by the witness, his evidence served only to demonstrate the fact, concurred in by all the parties, that there was a mistake somewhere in the plat. There is not enough land within the platted tract to supply all the lots of the dimensions given on the plat, and this situation is not controverted. Giving to Lot 22 a frontage of 75.38 feet results in moving each of the other lots twenty-five feet West, and in the end to completely extinguish or eliminate Lot 1. In other words, there is a deficiency of land, and all the lots cannot be accounted for. In such a case the most the court is authorized to do, in the form of correcting the apparent mistake, is to apportion the deficiency among the several lots, and not eliminate one of them entirely, as the trial court in effect did in the case at bar.

The owner of Lot 1 has as much, and it would seem a greater, right to have his property remain a part of the plat, as the owner of Lot 22; the greater right, because Lot 1 was first laid out by the owner of the plat, and beyond controversy, with the intention that it should be and remain a lot of the subdivision of the dimension indicated. The rule requiring an apportionment of either an excess or deficiency of land in such cases is well settled. . . . The trial court was therefore in error, even from the standpoint of the testimony of the witness Armstrong, in reaching the conclusion, in substance and effect that Lot 22 was entitled to remain with a frontage of 75.38 feet.

However, we are not to be understood as holding that the rule stated should have been applied to the facts in the case at bar. The rule applies more particularly to tracts of land subdivided into smaller tracts of specified and uniform dimensions, and not be a situation like that here presented. Here the owner of a definite tract of land intended, and his intention is manifest, to lay out as many lots of the uniform width of twenty-five feet as the tract would contain. This he proceeded to do, laying off twenty-one twenty-five foot lots, after which there remained an irregular or triangular piece to be disposed of as a remnant. This he supposed was 75.38 feet at its base, and so noted it upon his plat. This, we are satisfied from the record was a clear mistake. But it is not necessary to extend the opinion by a discussion of the matter, it is not important.

There is a deficiency of land to make up the number of lots with indicated frontage, and the rule in such a case is that the deficiency must fall upon the last or irregular tract; the remnant of the whole after laying out the lots of a uniform size. *Baldwin v. Shannon*, 43 N.J.L. 596. In a situation like this the owner of the plat must be deemed to have

intended to constitute the irregular remnant of a lot by itself, regardless of its dimensions, and a purchaser thereof takes the whole remnant, whether of greater or less area than that indicated by the plat. It cannot be enlarged at the expense of the owners of other lots, nor, if of greater area than shown by plat, diminished in their favor. Though the rule might in a given case work a hardship to the owner of the remnant lot, yet in the case at bar the combined injury to the owners of the other lots, arising, if defendant be sustained, from the displacement of their improvements and the total elimination of Lot 1, far outweighs any disclosed damage or hardship to defendant. The rule furnishes a definite and safe method and guide for the determination of mistakes of this nature, and we adopt and apply it to the fact in the case at bar; and if on a new trial the fact remains the same, the relief stated should be granted.

The decision of the state supreme court was correct, but the reason was wrong. In most states this situation would have been judged by the rule of a blunder: "Where a mistake occurs, leave it in that place." In this case, the shortage could be explained by scaling on the map. The following situation illustrates how a surveyor may be trapped by blindly following the remnant rule.

Shown in Figure 12.14 are the conditions found to exist along the former city boundary of San Diego. Blocks east of the boundary are all 4.8 feet long, more or

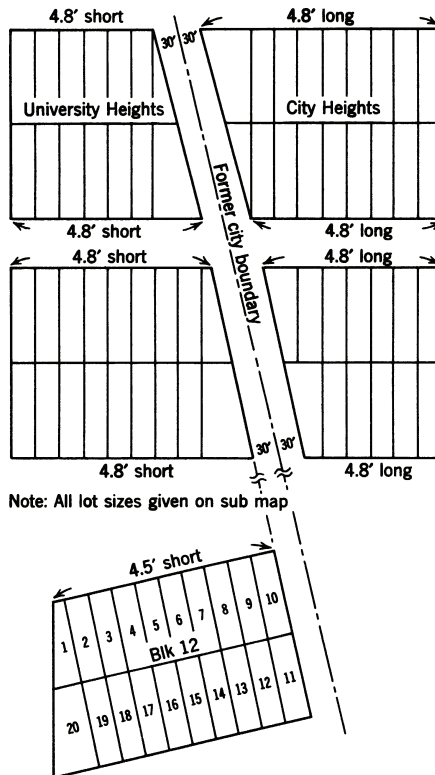


Figure 12.14 Conditions along the former San Diego city boundary.

less; blocks west of the boundary are all 4.8 feet short, more or less. Evidence shows that at the time Boundary Street was improved, the street improvements were located 4.8 feet too far to the west. The shift in position seems to have come about because when the line was run between monuments several miles apart it was found that the line was not straight. The 4.8-foot excess and the 4.8-foot deficiency belong to the remnant lots in Blocks 1, 2, 5, and 6, as this is where the error does in fact occur. In outward appearance, the remnant rule is correct. But in Block 12, where all lots were fully dimensioned, the error should be in Lots 10 and 11 (the city moved the line) and not the remnants.

In restoring lost corners in sectionalized lands, the Minnesota court rejected the remnant rule and adopted the apportionment principle. Figure 12.15 shows the conditions in *Goroski v. Tawner*.³² In three adjoining lines, a deficiency of about 15 chains existed in the last half mile. These sections are on the northern tier of a township and according to original survey instructions they are remnants; that is, errors of measurement were placed in the last half-mile. Regular forties were created except for the top tier.

The east quarter corner of Section 1 was lost. In restoring the lost corner, proportional measurement was used and the concept of placing the deficiency in the irregular closing lot was rejected. Thus, within the same state inconsistencies of principles can be found. It can be argued that in this case the state was following federal rules because it was sectionalized lands. A state does not have to accept federal apportionment rules for lands wholly under its jurisdiction.³³ It may choose to accept or reject any federal rule for restoring lost positions that does not involve federal ownership. In Michigan, the rule of apportionment of deficiency between lots of a block must stand irrespective of the end lot being fractional except where the contrary is proved.³⁴ The majority opinion on the remnant rule is well expressed by Judge Cassoday.

Had the plat given the specific dimensions of each lot of the several lots fronting on Jefferson Street except Lot 1, and given no dimensions of that, then such absence of the dimensions of that lot would have evinced an intention that it should include whatever should be left after setting off the several lots of which the specific dimensions had thus been given, whether the same should be more or less, but where, as here, the specific dimensions of each and all of the several lots fronting on Jefferson Street are given upon the plat, and there is no lot in the block of which the specific dimensions are not thus given; there seems to be no substantial reason why such excess should be given wholly to one lot merely because its dimensions, as given upon the plat, differ from those of other lots.³⁵

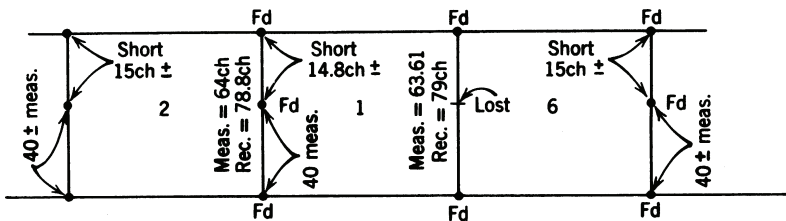


Figure 12.15 *Goroski v. Tawner*, 121 Minn. 189.

12.37 ESTABLISHMENT OF LOTS WHERE NO LOT MEASUREMENT IS GIVEN

Principle 29. In the absence of physical evidence on the ground and in the absence of measurements given on the plat, as a last resort scaling of the plat may be used.

Some early subdivisions failed to mention how big the subdivided lots were. On the map on which 10 lots, each scaling 50 feet, were shown, only 450 feet was found to exist on the ground. As several houses had been built by measuring 50 feet from each end of the block, a tangled legal snarl ensued. One of the unoccupied lots was finally purchased, and the block was resubdivided to adjust the houses within the land occupied. Scaling in this case indicated that there were 10 lots of equal size, about 50 feet each. Facts on the ground showed the existence of only 450 feet, or 45 feet per lot. Scaling could not be relied on to establish any fact other than that the lots were of equal size; the physical evidence on the ground was better evidence of the intent of the lot sizes.

Paper shrinks or expands with changing moisture conditions. A map reproduced by the dry process or the wet process will give a print different in scale from the original. A map drawn on a scale of 1 inch equals 100 feet cannot be scaled sufficiently accurately to determine whether a distance intended to be 50 feet is either 48 or 52 feet. Because of these facts, scaling must be considered poor evidence and is to be relied on only as a last resort.³⁶

The size of the drafting medium (paper) as well as its composition can also affect shrinkage and expansion, depending upon conditions. Extreme changes in moisture and temperature may produce dramatic changes.

12.38 ESTABLISHMENT OF LOTS WITH AREA ONLY GIVEN

Where a subdivision shows area only, the proration of any excess or deficiency presents almost insurmountable difficulties. Area is a computation depending on two linear measurements and directions, any of which can be altered. Sometimes area errors can be localized by the existing streets or lines of possession. Location of property lines from area data is considered a last resort to be applied after all other means have failed. "Where in an instrument of conveyance of real property the quantity of land to be conveyed is stated, the statement is not to be viewed as a factor in establishing boundaries unless the more particular description given is indefinite and uncertain."³⁷

12.39 NEW YORK RULE FOR ESTABLISHMENT OF LOTS

In New York and some other states, the custom has been to describe land by both lot and block in a subdivision and by a metes and bounds description. If excess or deficiency is found in the block, there is ambiguity, as two locations can exist. One

is based on proportional measurements and the other on record measurements from one end of the block. By the New York rule, if both descriptions are used, the first buyer gets exactly the dimensions called for (by metes and bounds and by the map) and later buyers can receive only a remainder (see Figure 12.16).

The facts in this case are undisputed. Both plaintiffs and defendants derive their title from a common source. On June 15, 1886, Ann C. Morton was the owner of certain property shown on a map, entitled “Building Lots for Sale at Middle Village.” This property was situated in block VI, as shown on the map, and included lots 1, 2, 3, 4, and 5, and also lots 26, 27, 28, 29, and 30 in that block. On that day she conveyed to one Jacob H. Gebhard said lots 1, 2, 3, 4, and 5, describing them by lot numbers according to the map, and also by metes and bounds, commencing “at a point on block VI formed by the westerly side of Morton avenue where it intersects with the southerly side of the Williamsburg and Jamaica turnpike road.” The description then continues along the turnpike road 125 feet, then southerly 107 feet 6 inches, then easterly 125 feet, to Morton Avenue, and “thence northerly, along the westerly side of Morton Avenue, 111 feet, 3 inches to the point or place of beginning.”

On February 14, 1898, Ann C. Morton conveyed to Joseph Burmel lots 26, 27, 28, 29, and 30, on the corner of Morton Avenue and Steuben Street, in block VI on the map in

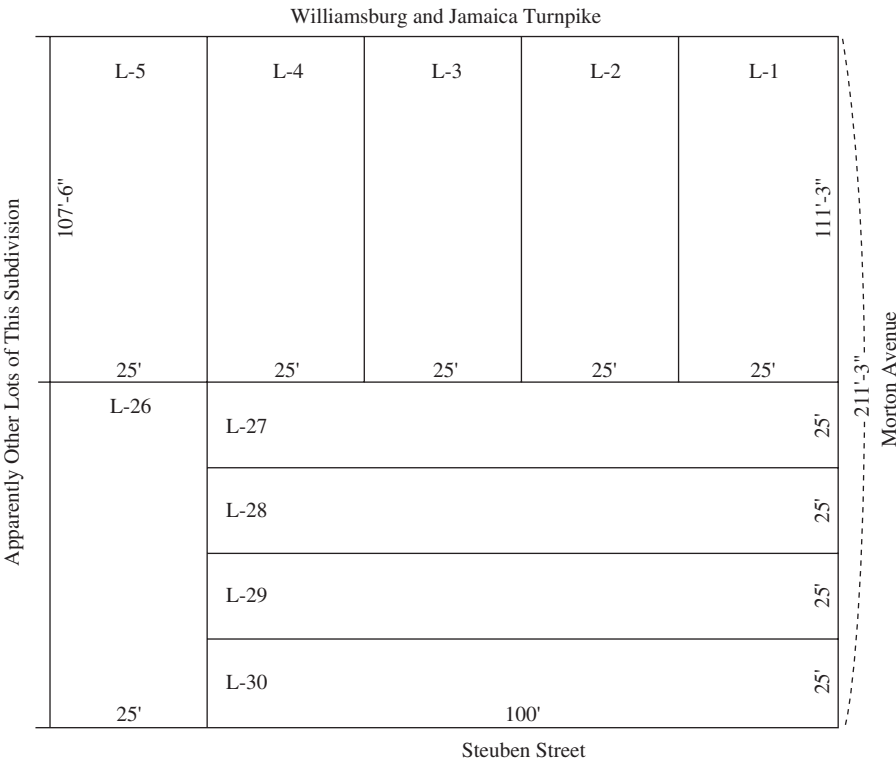


Figure 12.16 *Mechler v. Dehn*, 196 N.Y.S. 460 (1922).

question, reciting that said lots contained 12,500 square feet, more or less, and that said lot is shown on a diagram contained in the deed, "and agreeable to said map," but no such diagram appears. This deed contains no description by metes and bounds.

It appears from actual survey, and is undisputed, that the distance between the Williamsburg and Jamaica turnpike (now called Metropolitan Avenue) and Steuben Street, shown on the map as 211 feet 3 inches, is in fact only 206 feet and about 3 inches; in other words, that the actual distance between the turnpike and Steuben Street is 5 feet and 7/8 of an inch short of the distance shown on the map. The defendants now occupy the strip in dispute.

Upon these undisputed facts the trial court found that the description of the plaintiff's boundary on the westerly line of Morton Avenue as 111 feet and 3 inches was erroneous, and was only 106 feet 2 inches; that defendants were the owners of the remaining 100 feet on the westerly side of Morton Avenue; and that they did not unlawfully enter into possession of the plaintiff's premises.

The learned trial court states the general rule to be that where land is conveyed by reference to a plan or map, and there is more or less in the entire tract which has been divided than the map shows, no grantee is entitled to any preference over the others, and the excess should be divided among, or the deficiency borne by, all of the lots in proportion to their area, citing numerous cases from other jurisdictions. But he says that there is an exception to this general rule: that where a map shows a plotting of a considerable tract, and the creation of lots of regular width and depth, if a few of them are of irregular dimensions, they are deemed to be the remnant of what remained of the entire tract after plotting the regular lots, and if there is a shortage in the entire frontage, the irregular sized lots must bear it: that this is held upon assumption that the owner intended to get as many regular sized lots as possible, and that whatever remained of his frontage was to go into the irregular-shaped plot, and hence, if he had more or less than the map showed, the difference in the frontage would affect only the irregular lot, citing *Baldwin v. Shannon*, 43, N.J. Law, 596, and *Barrett v. Perkins*, 113 Minn. 480, 485, 130 N.W. 67.

Following this exception to the general rule as stated in his opinion, the learned trial court found that the entire deficiency should be taken from the plaintiff's lots, because they constituted the irregular lots in Block VI. There appears to be no decision in this state which precisely covers the situation presented in this case at bar, nor do the ordinary rules which have been adopted in this state seem to have any direct application.

I am unable to agree with the learned trial court in his final conclusion, which adopts the rule of law as to irregular lots laid down by the courts in New Jersey and Minnesota. I do not believe that this rule is sound in principle. A party who maps a plot of land into lots, whether regular or irregular, in size, and gives them certain dimensions on the map, believed by him to be accurate, cannot, in my opinion, be held by any such strained construction to intend the irregular lots as a remnant, regardless of its dimensions. On the contrary, I think he believes and intends that each lot shown on the map, regular or irregular, has, or should have, the dimensions ascribed to it. Especially is this true, in my opinion, where the so-called irregular lots constitute, as in the case at bar, approximately half the block. A literal application of this doctrine might result, in case of a large deficiency, in the loss of all, or nearly all, of the irregular lots, although held under a conveyance prior to that of the regular lots.

In the case at bar the original grantor clearly intended to convey in the deeds to plaintiffs' and defendants' predecessors in title 211 feet 3 inches of land fronting on Morton Avenue, 111 feet 3 inches to plaintiffs' and 100 feet to defendants' grantor. Her grantees intended likewise, in accepting such conveyance, to secure that amount of land. Had there been no map, concededly, as stated in the opinion of the trial court, plaintiffs' predecessor in title, having the first conveyance, would obtain title of the entire 111 feet 3 inches. How, by any legal principle, can the mere existence of the map and reference thereto, in the descriptions of the deeds, affect plaintiffs legal right, except by arbitrarily correcting the mistake and apportioning the deficiency, without any sound legal reason for such apportionment.

I do not think we are at liberty to presume from the mere reference to the map in the descriptions that the grantor intended to convey those lots in accordance with the frontage shown on the map, if it was there, or, if not, then in ratable proportion. I am unwilling to carry legal presumptions to any such extent. In my opinion, the mere statement of such a proposition shows its unsoundness.

The use of the words "more or less" cannot operate to modify or alter the grantor's intention, shown by the map and the descriptions. It certainly could have no such effect upon the first conveyance to plaintiff's grantor, because, measuring from the point of beginning, the lots as laid out on the map contain the frontage called for by the deed, and that amount of frontage existed and had not then been conveyed. When the subsequent conveyance of the remaining frontage on Morton Avenue was made, the grantor intended, of course, to convey 100 feet, believing that that frontage remained. But it is equally certain that she could not, and did not then intend to, convey more than she had. Therefore she conveyed to the defendants' grantor only what was left, or about 95 feet. I do not see that the fact that the lots are shown on a map and their description in the deed made by reference to that map in any degree alters this result. To correct the original mistake by an apportionment of the deficiency would be a purely arbitrary determination by the court, which could in no sense be called judicial.

Plaintiffs' grantor had a right to rely on the description in his deed and on the map, and also on the fact that, as this was the first conveyance on that frontage, the entire 111 feet 3 inches called for by his deed was there on the ground. No survey was necessary to determine that fact, nor was he bound to know that there was a deficiency in the block. On the other hand, defendants' grantor knew when he received his conveyance that he could obtain only what was left. He could then do one of two things, either determine for himself the correctness of the map and description by a new survey, or accept the deed and rely on the covenants against his grantor for damages for any deficiency. He did the latter. Why, then, should plaintiffs now be called on to make up any portion of this deficiency? In my opinion, therefore, the judgment should be reversed, and judgment directed for the plaintiffs for the relief asked for in the complaint.³⁸

In these results, the court recognized certain principles: (1) that proportionate measurement is a rule of last resort and (2) that the metes and bounds description and the senior deed indicated who should receive a *remainder*. In other jurisdictions, the courts reach as far as possible to discover who should receive the remainder and thus relegate proportionate measurements to a rule of last resort. Although the court could have invoked the remnant rule in this case, it did not (the results would have been reversed).

12.40 SUMMARY OF PRORATION RULES

From these situations and from others, the following general principles have evolved and can be relied on by the retracing surveyor:

Proportionate measurement is used to distribute discovered excess or deficiency as compared to the record where:

1. A whole is divided simultaneously into dimensioned parts by protraction; that is, the parts platted are not monumented on the ground, and no part is designated to receive a remainder. (See remnant rule.)
2. Equal parts are created by a will or partition; that is, each child receives a proportion of a whole.
3. Proportionate parts are created by the conveyance; that is, the west one-half and the east one-half of Lot 12, Map 1313.
4. To restore a lost monument position as nearly as may be to its original position where there are tie-outs from several directions, all of equal value (not applicable in sequence conveyances).

Proportionate measurement cannot be used:

1. To alter senior rights or in sectionalized lands to alter the superiority of one line over another.
2. To distribute a mistake that can be proved to exist in one place, that is, the rule of a blunder.
3. To alter legal rights obtained by possession.
4. To alter an acceptable original monument position.

Proportionate measurement:

1. Begins and ends at the nearest original monument position on each side of the area of excess or deficiency.
2. Is a rule of last resort.
3. Varies in interpretation in different jurisdictions.

12.41 ESTABLISHMENT OF LOTS ADJOINING SUBDIVISION BOUNDARIES

The foregoing discussion has been confined to lots within a subdivision but not adjoining the boundary of a subdivision. Lots abutting a subdivision boundary frequently require special attention where there was an error in the establishment of the original boundary line and at a later date the true boundary was recognized. The question of adverse rights often enters into these cases.

12.42 ESTABLISHMENT OF LOTS ADJOINING A SUBDIVISION CORRECTLY ESTABLISHED

Principle 30. Where the boundaries of a subdivision were staked correctly originally, the foregoing principles for the establishment of lots are applied

If there are no serious errors within the subdivision and corner monuments are found where they are indicated then one can rely on retracing the various lots within the subdivision based on the conclusive found evidence, which, by law, is without error.

12.43 ESTABLISHMENT OF LOTS OVERLAPPING THE TRUE SUBDIVISION BOUNDARIES

Principle 31. Lots abutting on a subdivision boundary line cannot extend beyond the title interest of the original subdivider except for a right of title resulting from lawful possession.

This principle is best explained by the following illustrations. The map of Middletown Addition filed in 1871 shows a picture plan and a general note giving the street widths, block sizes, and lot sizes. On the original map, the boundary line of the subdivisions was shown as in Figure 12.17. When the boundary was run out from known existing monuments on the ground, about 12 feet of the subdivision was not

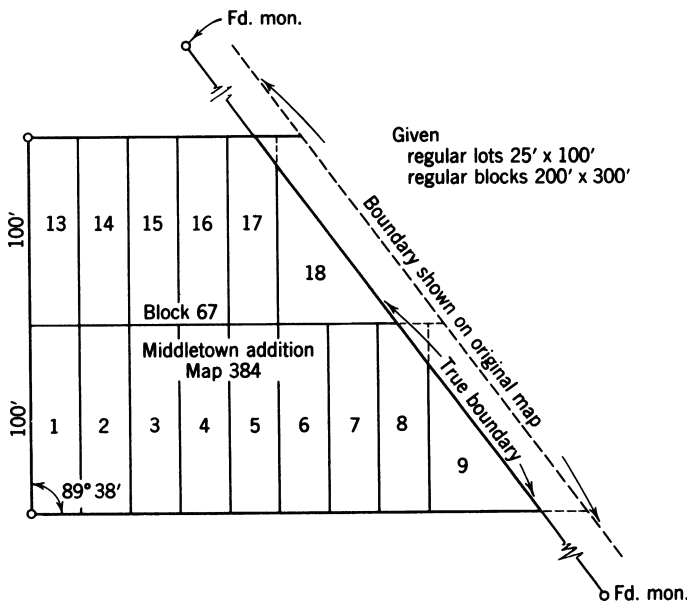


Figure 12.17 Subdividing another’s land in error.

in existence. Lots 8, 9, 17, and 18 cannot extend beyond the true boundary because the subdivider did not own the land and could not extend beyond his ownership, nor can the deficiency be distributed among all the lots in the block, as the original lot lines are unchangeable. The proper procedure in this block is to give each lot in the block 25 feet and place all the excess or deficiency in the lots adjoining the subdivision boundary.

Limitations on the Principle

As noted previously, title to land cannot be transferred by estoppel, adverse rights, agreement, and other unwritten means; without a written instrument, only the lines are affected.

12.44 ESTABLISHMENT OF LOTS NOT TOUCHING THE TRUE BOUNDARY OF THE SUBDIVISION

Principle 32. Where the original subdivider failed to subdivide all the land as shown by found original monuments, the surveyors should not extend the lot lines beyond the limits of the original monuments. Title to the unsubdivided land probably remains in the original subdivider.

When surveying College Park Unit 2, Map 2218, the surveyor failed to subdivide all the land owned by the subdivider by the amount shown in Figure 12.19. The original stakes for Lot 6 were found, and no doubt existed as to the original location of Lot 6. The fact that there was a strip of land varying in width from 10.6 to 12.94 feet, not subdivided by the subdivider, does not give Lot 6 a title interest to the strip (Figure 12.18).

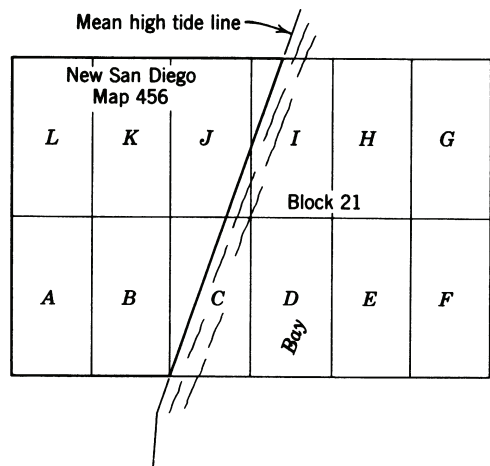


Figure 12.18 Land below the high-tide line cannot be privately owned.

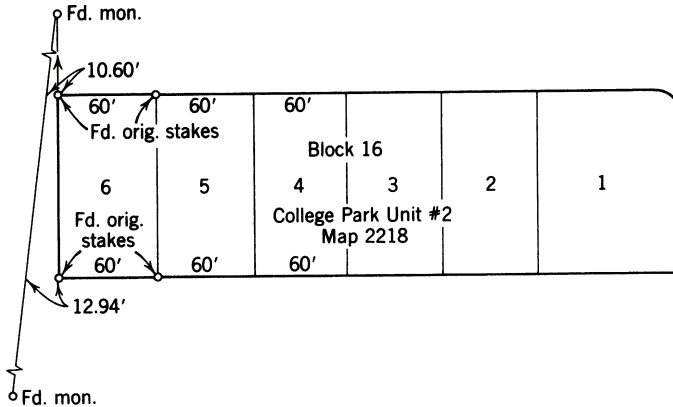


Figure 12.19 Gap between subdivisions.

Limitations on the Principle

This rule cannot apply to small, insignificant errors or cases where the original subdivider's monuments cannot be found. When a subdivider indicates by the title that all the land was subdivided and no monuments on the ground indicate otherwise, it must be presumed that all the land was intended to be subdivided.

In Washington, 4½ feet of surplus was found to exist between the street and the subdividers' true boundary line. The court ruled that when the lots, as marked on the plat, occupy the entire space between the north and south boundaries of the tract, the excess must be apportioned among the lots.³⁹

12.45 PRORATION OF EXCESS AND DEFICIENCY IN BLOCKS CLOSING ON SUBDIVISION BOUNDARIES

Principle 33. *A considerable excess or deficiency existing in blocks adjoining a subdivision boundary is prorated only as a last resort.*

Considerable error in the closing block of a subdivision may indicate an error in the original location of the subdivision boundary. If an error existed in the original boundary location of the subdivision, and if at a later date the boundary was moved to fit the true deed location of the subdivider, it is not advisable to move the lots by proration to fit the new boundary. Lots once established are unalterable.

There must be reasonable proof showing that an error probably existed in the original subdivision boundary location. If there are 10 blocks in a subdivision, 9 of which measure very close to the original record and the tenth, existing next to the subdivision boundary, is found to be 30 feet short, the inference is that the original boundary line was established erroneously and later moved to the true location. However, if the subdivision consists of only one block and the block is found to be 30 feet in error, proof is lacking to show on which end of the block the error might

have occurred, and in such a case the only equitable solution is to prorate the error. Blocks abutting on subdivision boundaries cause no end of grief to surveyors and landowners. Although a lot within a block that adjoins a subdivision boundary may be insured by a title company without fear of liability provided that it is described by lot number instead of size, the location and size of such a lot on the ground may be in serious doubt. Two possible solutions exist: (1) Prorate the error or (2) give the error to the end lot. If the error was brought about by a relocation of an incorrect original boundary line, the error would be applied where it occurred, that is, next to the boundary line. But if the error was not due to the boundary line being moved from its original location and no evidence exists to localize the error in one lot, the error would be prorated. This is a question of proof, the burden of which is placed on the surveyor.

12.46 LOCATING LOTS FROM BOUNDARY LINES

Principle 34. *It is safer to locate lots from the interior of a subdivision than from the boundary line of a subdivision.*

Proved interior lots and street locations within a subdivision are safer starting lines for staking the lots within a subdivision. The frequency of error in locating boundaries is far more common and is dangerous to use. Any lot or street within a subdivision should be located from the data on the map itself and not from a relocated boundary line of an adjoining subdivision.

Limitations on the Principle

In rare instances, the original lot lines of some of the early subdivisions were not staked on the ground, and in such cases the lots can be relocated only from the boundaries of the subdivision.

12.47 OBLITERATED AND LOST SUBDIVISIONS

Even after understanding all the preceding principles pertaining to boundary location procedures, and even after extensive practice in their application, the surveyor will find numerous perplexing situations created largely by conflicting evidence. Most surveyors can agree on measurements; they are either right or wrong. But when it comes to evaluating witness evidence, old documents, and physical evidence, many theories evolve. The mere fact that there have been thousands of court cases pertaining to the meaning of deed terms is sufficient proof that variations in interpretation occur.

In most cities, there are numerous subdivisions of lots and blocks. If the city is old, the chances are that the original subdivision data are not available. No one can say with any degree of certainty what was originally set and usually no one

can remember who the original surveyor was. The mere loss of original monument positions cannot and does not void a land title. Land is located as based on the best evidence available. In older areas, custom, occupancy, tradition, and related maps are often the only method available for locating lot lines. Surveyors with years of experience have background knowledge that tells them what is right. They are sought out by attorneys not because of their superior measurement methods but because of their firsthand knowledge of what has been. When retracing lots within ancient subdivisions, a new surveyor should do so with caution. Until the surveyor understands the unwritten history of similar subdivisions, he or she may not possess the qualifications to undertake such a task. The obligation of the surveyor is not to correct any errors made by the original surveyor; the obligation is to locate lot boundaries in conformity with what was actually done, and then to report the findings.

PROCEEDINGS IN PARTITION

12.48 GENERAL COMMENTS

A court decree resulting from proceedings in partition vests title in the various parties at the same time and by the same decree; none can be said to be senior to the other. Unless there is a definite statement limiting a party to a definite quantity of land, all are presumed to have a proportionate share of any surplusage or shortage.

12.49 ESTABLISHMENT OF LINES DETERMINED BY PROCEEDINGS IN PARTITION

In many proceedings in partition, a map or plan is made showing the division between the various allotments, and where the allotments are described by the map, the division lines between the various parcels are determined by the same principles as given previously for subdivisions. The map is treated like any other filed or recorded subdivision map. If the allotments are described by written language and there exists a gap or overlap between the allotments, the gap or overlap is prorated among the allotments.

The procedure for determining boundary lines resulting from proceedings in partition is clearly stated:

First: If the monuments or marks on the bounds for the corners of the several allotments can be found, such marks or monuments must govern, and distance and bearing must be disregarded.

Second: If the monuments or marks on the ground are lost or obliterated, parol evidence may be introduced in connection with the record to show their location.

Third: If no monuments were set, except theoretically or on paper, the proper location of these monuments will be determined by prorating the distances as given in the records, according to the length of frontage of the several allotments.

Fourth: If the actual computed sum of the lengths of the several allotments given exceeds the length of the tract partitioned, it will be construed that the decree means that, upon the hypothesis that the entire length of the whole tract is as stated, then the length of each assignment shall be (proportionately larger than) as given; but if it be less, the assignments of allotments must lose in like proportion.⁴⁰ Sometimes there is no map; however, the rules still apply.

12.50 ESTABLISHMENT OF BOUNDARIES OF ALLOTTEES OF WILLS

All heirs receiving land under a will have equal standing, and unless there are contrary words, each is to share in any surplusage or shortage. Where by the terms of a will the testator intended to give all her land to her heirs, and the tract is found to contain excess as shown by the sum of the acres given to the different heirs, the surplus will be prorated among the heirs proportionate to the named acreage of each.⁴¹ Again, usually there is no map, but the same rules still apply.

12.51 DEED DIVISIONS

Sometimes two or more people owning jointly would divide a tract of land by deed, describing their respective shares in a single deed, called a *division deed*. Unless there is a statement giving one party a remainder, the parcels so created are created simultaneously.

In many instances, when parties ask for an equitable division of land, they believe the court has the authority to subdivide a parcel by area. But in the strictest legal sense, a court of equity will ask that the land be sold and then the division of the prorated shares be distributed in the correct percentages.

12.52 COMMENTS

We *do not* consider the information contained in these chapters as legal advice for surveyors, but if placed in the hands of attorneys this information may be helpful. Sandwiched between many complex survey concepts the surveyor will find legal doctrines with which each surveyor should become familiar as being necessary to practice boundary creation and retracement.

This book encourages each and every surveyor to become acquainted with surveying principles that are necessary to perform his or her work and let his or her coworker, the attorney, make the legal conclusions.

NOTES

1. *O'Brien v. McGrane*, 27 Wis. 446 (1871).
2. *Rivers v. Lozeau*, 529 So. 2d 1147 (Fla. App. 1989).
3. *Pyburn v. Campbell*, 158 Ark. 321 (1923).